

GSFC NEON Series-1 CMO

RELEASED

NS1-SMBA-CDRL-0001, Revision C
Near Earth Orbit Network Series-1 Project, Code 473

**Near Earth Orbit Network
Series-1 Project
Sounder for Microwave-Based
Applications
Contract Data Requirements List**



**Goddard Space Flight Center
Greenbelt, Maryland**

**Near Earth Orbit Network Project Series-1
Sounder for Microwave-Based Applications
Contract Data Requirements List
Review/Signature/Approval Page**

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Preface

This document is under NEON Series-1 (NS1) Project configuration control. Once this document is approved, NS1 approved changes are handled in accordance with Class I and Class II change control requirements as described in the Low Earth Orbit (LEO) Programs Division (LPD) Programs and Projects Configuration Management Procedure (CMP), and changes to this document shall be made by complete revision.

Any questions should be addressed to:

LEO/NEON Configuration Management Office
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Change History Log

Revision	Effective Date	Description of Changes
Revision –	December 20, 2024	This version incorporates 470-CCR-24-0404 which was approved by the SMBA ERB on March 26, 2024, and incorporates NS1-CCR-0004 by the NS1 SMBA CCB on the effective date shown.
Revision A	April 1, 2025	This version incorporates NS1-CCR-0007 which was approved by the NS1 SMBA CCB on the effective date shown. Removed duplicate language for UAR. Changed delivery date for MA-1.
Revision B	August 19, 2025	This version incorporates NS1-CCR-0014 which was approved by the NS1 SMBA CCB on the effective date shown.
Revision C	January 15, 2026	This version incorporates NS1-CCR-0017 which was approved by the NS1 SMBA CCB on the effective date shown.

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1 INTRODUCTION

This contract data requirements list (CDRL) document defines the requirements for deliverable documentation to be provided by the Sounder for Microwave-Based Applications (SMBA) Contractor. Section 1 includes definitions and instructions for distribution. Table 1 presents the CDRL item by item, with due dates, quantity, and media format. Section 2 contains the data item descriptions (DID), descriptions of each item which provide the use for the deliverables and the required preparation information. Except where specifically indicated to the contrary, the formats and drawing standards used **shall** be those normally used by the Contractor and by its subcontractors.

1.1 Scope

The CDRLs within this document include those required for the development phase as well as the production phase, as indicated in Table 1. Though this document has been created for the Near Earth Orbit Network (NEON) Series-1 (NS1) project, the CDRLs, particularly those associated with the production phase, are applicable to subsequent projects within the NEON program.

1.2 Export Control Limitations

For Export Control limitations reference the Export License clause in the Contract.

1.3 Proprietary Data

For Proprietary Data requirements reference the Rights in Data clause in the Contract.

1.4 Prior Submission Deliveries

If new hardware, software, or documentation is developed, procured, designed, fabricated, tested, or processed in a different manner from that of the first SMBA flight Instrument effort (obsolescence, corrective actions, modifications, errors, etc.), the Contractor **shall** deliver the data item as required with the necessary updates for Government approval, reference, or information, as defined for each impacted deliverable herein.

1.5 Due Dates – Related Definitions

The following definitions apply to the "DUE DATE, MATURITY" column in Table 1. Unless otherwise specified, deadlines are in calendar days.

- DUE DATE:
 - CDR, PER, etc.: Critical Design Review (CDR), Pre-Environmental Review (PER), Pre-Ship Review (PSR), etc. (all Top-Level Instrument Reviews).
Electronic distribution to be delivered to the Government five working days prior to review, unless otherwise stated.
 - Monthly: Submitted on monthly basis
 - L: Launch Date (Actual Launch Date)
 - DACA/(D): (Calendar) Days After Contract Award/(Definitization)

- ORR: Mission-Level Operational Readiness Review
- PLAR: Post-Launch Assessment Review (at the completion of on-orbit commissioning)
- MATURITY:
 - Initial: The first submission of an item which is anticipated to be revised and resubmitted at a later date.
 - Preliminary/Prelim: An early submission of an item. To be completed as is practicable at the time of preparation. Preliminary submittals are written with the best available current information and are resubmitted when further information becomes available.
 - Final: The complete, thorough submission of an item for approval, review, or information that, to the best of the Contractor's knowledge and intention, will not require further revision or updates. However, this does not preclude updating if later found to be necessary. Any updates **shall** require the same "approval/review" process as was required for the original submissions.
 - Current: The delivery is written with the best up-to-date information available at the time.
 - Update: The delivery is revised with the best up-to-date information available at the time.

Other entries in the "DUE DATE" and "MATURITY" columns are self-explanatory.

1.6 Media – Related Definitions

The following definitions apply to the deliverables in Table 1.

E – Data items **shall** be delivered in electronic format to a GSFC Low Earth Orbit (LEO) Programs Division (LPD) specified web portal unless otherwise noted in the CDRL DID. The Contracting Officer **shall** be notified of electronic submission of the deliverable via email. Electronic deliverables **shall** be delivered in the following formats unless otherwise approved by the Government:

Text Documents:	PDF (searchable) or Microsoft Word
Presentations:	PDF (searchable) or PowerPoint
Spreadsheets:	Microsoft Excel
Database:	Delimited ASCII files accompanied with database schema document defining tables and entries.
Schedules:	PDF and Microsoft Project

Schematics and Drawings:	Design Web Format (DWF) and PDF
Photographs:	JPEG or current industry standard.
Video:	Any readily available open standard (e.g., AVI, MPEG) Test
Measurement Data	Delimited ASCII format or other acceptable format

In addition to deliveries in the required formats above, the Contractor may also provide additional copies of the same deliverables in other formats (e.g., .mdzip).

If separate distribution instructions are not specified for separate submission due date/maturity items, then the listed distribution applies to all submissions.

1.7 Category – Related Definitions

The following definitions apply to the "Submission Category (CAT)" column in Table 1. If separate approval instructions are not specified for separate submission due date/maturity items, then the listed approval instruction applies to all submissions.

- A Approval: CDRL items in this category require approval by the GSFC COR or Contracting Officer prior to use by the Contractor. Receipt by the Government **shall** occur within the time specified in the "Due Date" column of Table 1 of this document. Requirements for resubmission **shall** be as specified by the Contracting Officer. For most cases the Contractor **shall** be required to resubmit the document within 14 calendar days of receiving comments from the Government. If the Contractor has not received response from the Government within 30 calendar days of delivery of a CDRL item, the Contractor may proceed as if the document has been approved.
- I Information: Data in this category **shall** be delivered to the Government within the time period specified in the "Due Date" column of Table 1 for the purpose of determining current program status, progress, and future planning requirements.

1.8 Applicability

The "Applicability (APPLIC)" column in Table 1 indicates the Contract Line Item Numbers (CLIN) for which each CDRL is applicable and during which the CDRL is to be delivered. This is done by indicating Development [DEV], Production [PROD], or Task Order [TO] CLINs. CLINs are associated with Development [DEV], Production [PROD], or Task Order [TO] per the Contract Clause GSFC 52.211-91, SCOPE OF WORK.

1.9 Updates

If the Government or Contractor determines that there is a Contractor-initiated substantive change and an update is needed to properly reflect the information previously delivered, the Contractor **shall** make the update and redeliver the CDRL independent of the quantity requirements in the CDRL table. The Contractor **shall** notify the Government of any pending updates and re-deliver the CDRL on a schedule agreed to by the Government. Table 1 does not specify this repetitious requirement; however it is required for each CDRL.

1.10 Documentation Change Procedures

Unless otherwise defined in the Statement of Work (SOW) or contract:

- When a minor correction/change is needed or has been made to a few parts of a “Final” version of a CDRL item that has already been delivered to the Government, the Contractor **shall** issue within 30 days a Documentation Change Notice starting with the number 001. Change bars or “change tracking” **shall** be used to indicate the locations of changes unless the particular file type does not have those features available. If not available, the details of the changes will be described in a text file.
- When a major correction/change is needed or has been made to all or multiple parts of a “Final” version of a CDRL item that has already been delivered to the Government, the Contractor **shall** deliver within 30 days a complete revision of the CDRL item. A description of all changes incorporated into the new revision **shall** be included in the new document or otherwise documented. No change bars are used when a document is updated by revision, and the Documentation Change Notice numbers for that document **shall** be automatically recycled to 001.

1.11 Delivery Instructions

CDRL submissions **shall** be complete, unabridged, and unredacted. All CDRLs **shall** comply with the contract rights in data and export control clauses.

External documents referenced within a CDRL submission **shall** be made available virtually to the Government upon request.

The Contractor **shall** deliver each CDRL item via the media indicated in accordance with the schedule and any special instructions indicated in Table 1 and/or the corresponding DID, and in accordance with the contract or SOW requirements and provisions.

All CDRL items **shall** be delivered electronically (one electronic copy for each CDRL item) to an electronic library designated by the Government. If all or a portion of a CDRL item is not well-suited for electronic delivery, the item may be delivered via hardcopy upon mutual agreement between the Contractor and the Government.

All CDRL submittals **shall** be protected by encryption or through the use of other secure means approved by the Government against unauthorized disclosure including interception and decryption by foreign nationals. Prior to delivery, the Contractor **shall** implement all safeguards necessary to ensure data is protected in accordance with the Export Administration Regulations except when expressly authorized by a Technical Assistance Agreement.

The Contractor **shall** include a cover letter with the following information:

- CDRL #
- Title
- Due Date
- Flight Model
- Serial Number
- Original Submission Date

- Prior Delivery CDRL #
- Current Submission Revision
- Current Submission Revision Date

The Contractor may deliver a single letter for multiple submissions as long as the information above is contained in the letter for each deliverable item.

Table 1. Contract Deliverable Items List

No.	MANAGEMENT	DUE DATE, MATURITY	CAT	APPLIC
PM-1	Monthly Project Management Reviews (PMR)	Monthly, 3 days prior to review	I	DEV, PROD, TO
PM-2	Engineering Peer Review (EPR) Plan	Initial: Development CLIN Kickoff Final: 60 DACA	I A	DEV
PM-3	Integrated Program Management Data and Analysis Report (IPMDAR)	Initial: Draft Integrated Master Schedule with Proposal Initial Baseline: 45 DACA Baseline: 12th working day of the 3rd fiscal month after contract definitization Monthly financials: by the 12th working day following the close of the prior month accounting period	I I A I	DEV
PM-4	Integrated Master Schedule (IMS)	Baseline: Production CLIN Kickoff Monthly IMS snapshot no later than the 5th working day of the month Monthly Final: By the 12th working day following the close of the prior month accounting period	I I I	PROD, TO

No.	REVIEWS	DUE DATE, MATURITY	CAT	APPLIC
RE-1	Kickoff Meeting Data Package	Final: 14 DACA of CLIN	I	DEV, PROD
RE-2	System Requirements Review (SRR)/System Definition Review (SDR) Data Package	Final: 5 days prior to review	I	DEV
RE-3	Preliminary Design Review (PDR)	Final: 5 days prior to review	I	DEV

RE-4	Critical Design Review (CDR)	Final: 5 days prior to review	I	DEV
RE-5	Pre-Environmental Review (PER)	Final: 5 days prior to review	I	DEV, PROD
RE-6	Pre-Ship Review (PSR) Data Package	Final: 5 days prior to review	I	DEV, PROD
RE-7	Integrated Baseline Review (IBR)	Final: 5 days prior to review	I	DEV
RE-8	Engineering Peer Review Data Packages	Engineering Peer Review (EPR) Meeting Material : EPR -3 days Final: Electronic or Hard copies at review	I I	DEV
RE-9	Manufacturing Readiness Review Package	Final: 5 days prior to review	I	DEV, PROD
RE-10	Test Readiness Review (TRR) Data Package	Final: 5 days prior to review	I	DEV, PROD, TO
RE-11	Inputs to Project Level Reviews	Final: 30 days prior to review	I	DEV, PROD, TO

No.	SYSTEMS ENGINEERING	DUE DATE, MATURITY	CAT	APPLIC
SE-1	System Performance Verification Plan (SPVP) and Matrix (SPVM)	Initial: SRR - 7 days Final: CDR - 7 days Update: PSR	I A A	DEV, PROD
SE-2	Trend and TPM Plan/Analysis, Reports, and Operations Log	Plan Initial: PDR Plan Update: PER Plan Final: PSR Analysis, Reports, and Logs delivered per the plan.	I A A	DEV, PROD, TO
SE-3	Engineering Drawings	Indentured Drawing List: Make Available Virtually at Top-Level Instrument Reviews	N/A	DEV

SE-4	Engineering Change Requests	Class I: As Generated	A	DEV, PROD, TO
SE-5	Instrument Performance Specification	Initial: SRR -7 days Update: PDR - 7 days Final: CDR - 7 days	I I I	DEV
SE-6	Structural and Mechanical Analysis and Models	Initial: PDR -30 days Final: CDR -30 days Correlated Model: PSR -30 days	I A A	DEV
SE-7	Thermal Analysis and Models	Initial: PDR -30 days Final: CDR -30 days Correlated Model: PSR -30 days	I I A	DEV
SE-8	Contamination Analysis Report	Initial: CDR Final: PER Update: PSR -30 days	I I I	DEV, PROD
SE-9	Electromagnetic Interference (EMI)/Electromagnetic Capability (EMC) Analysis Report	Initial: CDR – 7 days Final: PSR -30 days	I A	DEV
SE-10	Instrument Concept of Operations Document	Initial: SRR – 7 days Final: PDR – 7 days	I I	DEV
SE-11	Inputs to the Instrument to spacecraft Interface Control Document (ICD) including the Mechanical ICD	Initial: PDR Final: CDR	I A	DEV, PROD
SE-12	Field Programmable Gate Array (FPGA), Application Specific Integrated Circuit (ASIC), and	Initial: PDR -30 days Final: CDR -45 days	I I	DEV

	Programmable Logic Device (PLD) Circuitware Design Data Package	As generated for each FPGA/ASIC/PLD		
SE-13	Mass Properties Report	Final: PSR	I	DEV, PROD
SE-14	Spare Parts Plan and List	Final: Development CLIN Kickoff	A	DEV
SE-15	Engineering Development Plan	Initial: Development CLIN Kickoff Final: SRR- 7 days	I A	DEV

No.	SOFTWARE	DUE DATE, MATURITY	CAT	APPLIC
SW-1	Instrument Software Development Plan	Final: 30 DACA	A	DEV
SW-2	Software Detailed Design Document	Initial: PDR-90 days Final: CDR – 90 days	I I	DEV
SW-3	FSW Version Description Document	Final: 30 days after FSW acceptance test completion and with each FSW release/build	I	DEV, PROD, TO
SW-4	Instrument Software Maintenance Plan	Final: PSR	I	DEV
SW-5	Instrument Data Sets	Final: PSR	I	DEV, PROD, TO

No.	INTEGRATION AND TEST	DUE DATE, MATURITY	CAT	APPLIC
IT-1	Instrument Integration, Test, and Verification Plan	Initial: PDR -7 days Final: CDR -7 days FM2 Update: Updated version prior to FM2 I&T based on lessons learned from the FM1	I I I	DEV
IT-2	Test Results/Reports	Final: 10 days after test completion	I	DEV, PROD, TO

IT-3	Test Procedures & As-Run Test Procedures	Pre-Test: 10 days prior to test	A for Pre-Test	DEV, PROD, TO
		As-Run: Conclusion of each test + 7 days	I for As-Run	
		<u>Launch site-specific Test Procedures</u>		
		Initial: L -12 months	I	
IT-4	Instrument Post Delivery Support (PDS) Document	Final: L -3 months (i.e., SC PSR)	A	DEV
		Initial: PSR -120 days	I	
		Final: PSR -7 days	A	

No.	MISSION ASSURANCE	DUE DATE, MATURITY	CAT	APPLIC
MA-2	Orbital Debris Assessment Report (ODAR) and End of Mission Plan (EOMP)	Initial ODAR inputs: PDR -15 days	I	DEV, PROD
		Interim ODAR and Draft EOMP inputs: CDR -60 days	I	
		Final/updated ODAR and EOMP inputs: PSR -90 days	I	
MA-3	System Safety Plan	Initial: SRR – 7 days	I	DEV
		Final: PDR -45 days	I	
		Updates 30 days prior to implementation	I	
MA-4	Safety Requirements Compliance Checklist	Initial: SRR -45 days	A	DEV, PROD, TO
		Final: PDR -45 days	A	
		Updates: CDR -45 days	A	
MA-5	Instrument Safety Assessment Report	Initial: PDR -30 days	A	DEV, PROD
		Intermediate: CDR -30 days	A	
		Final (with VTL): PSR -45 days	A	

MA-6	Failure Modes, Effects, and Criticality Analysis (FMECA) and Critical Items List (CIL)	Initial: PDR -30 days Final: CDR -30 days	I I	DEV
MA-7	Master EEE Parts List	EEE parts to the Parts Control Board prior to inclusion Updates prior to inclusion	I I	DEV
MA-8	Materials Identification and Usage List (MIUL)	Initial: PDR -30 days Final: CDR -30 days Updates: 30 days after identification	I I I	DEV
MA-9	Life Test Plan and Final Report for Lubricated Mechanisms	Plan: PDR -30 days Report: 30 days after acceptance test completion	I I	DEV
MA-10	End Item Acceptance Data Package	Component/Subsystem Final: EIDP for Critical Items (as identified in CDRL PM-2, EPR Plan) with delivery of the item Initial: PSR -30 days Final: 30 days prior to end item delivery Updates: 30 days after identification	I (for crit. comp/subsy) I A A	DEV, PROD

No.	CALIBRATION/VALIDATION	DUE DATE, MATURITY	CAT	APPLIC
CV-1	Calibration/Validation Plan	Initial: PDR -30 days Final: CDR -30 days	I A	DEV
CV-2	Calibration/Validation Procedures	Initial: 60 days prior to use Final: 10 days prior to use <u>On-orbit Procedures</u>	I A	DEV, PROD, TO

		Final: ORR -30 days Update: PLAR	A A	
CV-3	Calibration/Validation Results	Electronic Data: 3 days post-test Test Report: 10 days post-test Analyses: At completion + 10 days, NLT PSR	I I I	DEV, PROD
CV-4	Radiometric Math Model (RMM)/Sensor Performance Math Model (SPMM)	Initial: CDR -30 days Final: PSR -30 days	I A	DEV, PROD
CV-5	Calibration Data Book	Initial: CDR Final: PSR Update: after commissioning	I I A	DEV, PROD, TO
CV-6	Ground Processing Software	Initial: CDR -30 days Final: PSR -30 days	I A	DEV
CV-7	Spectral Response Function (SRF) Measurements	Final: PSR -7 days	A	DEV, PROD
CV-8	Hyperspectral Capability Test and Analysis	Initial: CDR Final: PER -30 days	I A	DEV
CV-9	Radio Frequency Interference Detection Capability Test and Analysis	Initial: CDR Final: PER -30 days	I A	DEV

No.	ON ORBIT	DUE DATE, MATURITY	CAT	APPLIC
OO-1	Instrument On-Orbit Commissioning Plan	Final: PSR -30 days	I	DEV
OO-2	Instrument User's Manual	Initial: PER Final: PSR	I I	DEV

OO-3	Commissioning Report	Initial: 10 Days after each test Test Results: PLAR Final: PLAR	I I I	DEV, PROD, TO
OO-4	Command and Telemetry Handbook and Command and Telemetry Database (CTDB)	Initial: PER -30 days Final: PSR -6 months	I I	DEV, PROD
OO-5	Instrument Launch and On-Orbit Operations and Contingency Procedures	Final: PSR	I	DEV
OO-6	Instrument Simulator Users Guide	Initial: CDR Final: with shipment of the OpSim	I I	DEV

2 DATA ITEM DESCRIPTIONS

2.1 Project Management Data Items

DATA ITEM DESCRIPTION

1. CDRL No.:

PM-1

2. Title:

MONTHLY PROJECT MANAGEMENT
REVIEWS (PMR)

1. Reference:

2. Use:

Serves as a primary means to enable project team insight into the SMBA contracted efforts and their status. These review packages will be used to both document status and various metrics with updates monthly and provide an opportunity for discussions between the Contractor and the Government regarding project status, plans, trends, risks, and issues.

3. Preparation Information:

The PMR **shall** include reporting across the full scope of contract effort including subcontractors'/suppliers' efforts as appropriate to report the information required per this CDRL DID.

The Monthly PMR **shall** include the following information in a format and organization defined by the Contractor:

- Program Manager's Assessment and Programmatic Overview (e.g., significant accomplishments, new/open/closed issues, significant upcoming events, significant changes in plans, etc.).
- Significant or impactful problems encountered during the reporting period, and anticipated approaches for resolution (including, as appropriate, technical issues including hardware, software, and Ground Support Equipment (GSE) issues, manpower and staffing, and supplier and subcontractor issues, etc.); status of open issues and problems from prior reporting periods.
- Risk status for top 10 risks (and all red risks) with mitigations plans and opportunities for each.
- CDRL status for each document delivered in accordance with the CDRL or overdue from previous reporting periods and status of documents planned for delivery during the next reporting period.

- Key personnel changes.
- Status of action items.
- Summary of Integrated Master Schedule
 - Schedule and schedule performance status for the contracted effort that includes detail for each subsystem/subassembly
 - The Primary, Secondary, and Tertiary critical paths with any potential mitigation opportunities that are being considered to maintain the schedule
 - Critical Path and subsystems schedule reserve/slack trend metrics as jointly defined by the Contractor and Government
- Control Milestone Trend Report – The Contractor **shall** provide a control milestone trend chart that consists of the baselined list of control milestones (determined by the Contractor and agreed to by the project office) and their baselined and projected or actual completion dates. The report will also contain a list of the control milestones expected to complete during the reporting period, their baselined and projected completion dates, and their current status. Note that changes to the control milestones **shall** be submitted as part of the monthly report for review and concurrence by the Project Office.
- Status of Giver/Receiver items (e.g., item, giver or receiver, need date, forecast availability, etc.).
- For Cost Plus CLINs, a summary explanation of key Earned Value Management (EVM) parameters performance and trends; summary of financial status and cost performance; status of management reserve, liens, or other unfunded risk mitigations including estimated cost.
- For Fixed Price CLINs, summary of Milestone Payments paid, unpaid/open and status, and a 12-month forecast of upcoming milestones.
- For Cost Plus CLINs, headcount metrics with variance explanations reported at the Integrated Product Team (IPT) and Total Project levels that summarize Baseline, Forecast, and Actual headcounts where a “head” equals one equivalent work-month of effort.
- Technical status, risks, and issues of the total contracted effort that includes significant/higher risk/near critical path subcontracts
 - Including a list of suppliers (e.g., significant suppliers/subs who provide higher risk items, items of significant value, or items near/on the critical path), description of product being supplied, and status
- Technical Performance Metrics (TPM) (reference CDRL SE-2) that show the Current Best Estimated (CBE), margin/contingency (adjusted for maturity) to the specification indicating calculated, and/or measured values, their accuracy, and units with trends for each including software metrics (reference CDRL SW-1).
- Integration and test (I&T) planning summary, test flow/schedules, reviews, procedure release status, readiness, significant completed/open work, risks, issues, concerns, significant lessons learned, and Verification and Validation status (e.g., burn down metrics) when available.

Backup Material:

- Spare parts status
- Configuration Control Board (CCB) Status: List of upcoming, in process, and approved Class I changes; list of approved Class II changes

- Mission Assurance Program Status
 - Summary of Material Review Board (MRB) actions and status
 - Summary of Failure/Anomaly Review Board (FRB/ARB) actions and status
 - Summary of Failure investigations and corrective actions
 - Company/program quality metrics relevant to processing and program
 - Manufacturing quality statistics for higher risk/higher impact subcontractors:
First pass yields; Supplier defect rates, etc.
- Prohibited Materials Compliance Status (as available)
 - SMBA Product Alert evaluation status
 - Production Readiness issues (as available)
- Complete risk list from database with risk statements, LxC, and status (open/closed/accepted)
- Project security status

DATA ITEM DESCRIPTION

1. CDRL No.:

PM-2

2. Title:

ENGINEERING PEER REVIEW (EPR) PLAN

3. Reference:

GPR 8700.6A, Engineering Peer Reviews

4. Use:

To communicate the EPR design review plan to the Government for early planning, discussion, and streamlined execution. It is also a means to convey which elements of the contracted effort are of higher criticality allowing early discussion and understanding. Lastly, it provides an early means to ensure the review approach will be focused on the engineering, streamlined, and not planned with inefficient overhead in chart development and unnecessary presentations.

5. Preparation Information:The EPR Plan **shall**:

- Identify major/critical subsystems to be reviewed. Critical subsystems are those components, subassemblies, subsystems, and software that are higher risk/complexity items, items of significant dollar/schedule value, or items near/on the critical path. Critical suppliers are those suppliers delivering critical subsystems.
 - Critical subsystems and suppliers identified in the EPR plan are referred to elsewhere in the SOW and CDRL where there is additional attention required for these items (ex. peer reviews, additional deliverables)
- Identify the processes, plans, components, subassemblies, subsystems, software elements, etc., that will be peer reviewed.
- Provide a schedule tied to major reviews/milestones for the planned EPRs.
- Identify components and subsystems that will not be peer reviewed and state the rationale.
- Describe the methodology used to determine the scope of the EPR process.
- Describe the planned peer review process, including notional review personnel, any plans for internal independent reviewers, nominal agenda, and action item management.

DATA ITEM DESCRIPTION

1. CDRL No.:

PM-3

2. Title:INTEGRATED PROGRAM MANAGEMENT
DATA AND ANALYSIS REPORT (IPMDAR)3. Reference:

Integrated Program Management Data Analysis Report (IPMDAR) Implementation & Tailoring Guide, 24 August 2021

(https://www.acq.osd.mil/asda/ae/ada/ipm/docs/IPMDAR%20Implementation%20and%20Tailoring%20Guide_Stamped.pdf),

Data Item Description for Integrated Program Management Data and Analysis Report (IPMDAR) DI-MGMT-81861C (this document is to be used as guidance in the preparation of the IPMDAR) [ASSIST-QuickSearch Document Details \(dla.mil\)](#)

Contract Performance Dataset (CPD) validator tool [EVM-CR Resources \(osd.mil\)](#)

Schedule Performance Dataset (SPD) validator tool [EVM-CR Resources \(osd.mil\)](#)

CPD FFS [20200312 Contract Performance Dataset - FFS.pdf \(osd.mil\)](#)

SPD FFS [20200312 Schedule Performance Dataset - FFS.pdf \(ssitools.com\)](#)

SP-2010-403, [NASA Schedule Management Handbook](#)

[NASA IPMDAR Tailoring Supplement \(NITS\)](#)

4. Use:

For Cost Plus CLINs, to provide information for: (1) integrating cost and schedule performance data with technical performance measures; (2) assessing the magnitude and impact of actual and potential problem areas causing significant cost and schedule variances; and (3) providing valid, timely project status information to higher management.

5. Preparation Information:

To ensure an integrated approach to project management, the data provided by this IPMDAR Data Requirements Document (DRD) **shall** be in consonance with the work breakdown structure (WBS), organizational breakdown structure (OBS), risk management processes, integrated master schedule (IMS), plans and reports (where required), probabilistic risk assessment processes and reports (where required), the cost analysis data requirement (CADRe) (when required), and the monthly/quarterly Contractor financial management reports (NASA 533M/Q). The Contractor must be able to reconcile between the NASA 533M/Q and the IPMDAR. This reconciliation should be included within the required IPMDAR performance narrative report.

IPMDAR contains data for measuring Contractor cost and schedule performance at NASA. The primary purpose is to reflect current performance status and the forecast of future performance.

The IPMDAR consists of the following components:

- Contract Performance Dataset (CPD) – Provides performance/execution data from Contractor existing management systems.
- Schedule (comprised of both Microsoft Project or Native Schedule File and the Schedule Performance Dataset (SPD) – Provides data from the consolidated Contractor IMS. At a minimum, subcontractor effort must be incorporated into the IMS at an intermediate level with key milestones and be distinguishable from the Contractor's and subcontractor's efforts.
- Performance Narrative Report (Comprised of both the Executive Summary and Detailed Analysis Report) – Provides narrative analysis of data provided in the CPD and the SPD.

The IPMDAR CPD and SPD files **shall** be validated prior to submittal.

The CPD and SPD formats will be JavaScript Object Notation (JSON) format as defined by the IPMDAR file format specifications (FFS) and the IMS in Microsoft Project file or native schedule format. The IMS data dictionary and performance narrative report will be in Microsoft Word/Excel or native format.

CPD, Schedule (Microsoft Project or Native Schedule File and SPD) and Performance Narrative Report (Executive Summary and Detailed Analysis) Instructions:

5.1 CPD Instructions: Contractor will meet the requirements set forth in [NASA IPMDAR Tailoring Supplement \(NITS\)](#)

The CPD **shall** be submitted in dollars and hours as part of the original JSON submission. Contractors with IPMDAR flow-down **shall** submit dollars and hours as part of the original Prime submission and provide a separate JSON file with dollars or hours directly to the Government to protect data sensitivities related to rates.

Data Repository-Electronic Submission and File Formats: Data in accordance with the applicable approved CPD FFS and Data Exchange Instructions is to include all structures and units to at least the control account/work package level

Subcontractor Integration: Data for Subcontractors with an IPMDAR DRD flow-down requirement **shall** be integrated with the Prime's IPMDAR CDRL delivery.

5.2 SPD: Contractor will meet the requirements set forth in [NASA IPMDAR Tailoring Supplement \(NITS\)](#).

The Contractor **shall** use Microsoft Project as their scheduling tool for integration with GSFC schedules.

Contractors with IPMDAR flow-down may submit dollars and hours as part of the original Prime submission and provide a separate JSON file with both dollars and hours directly to the Government to protect data sensitivities related to rates.

IMS Data Dictionary for Microsoft Project or Native IMS Schedule File: The first monthly analysis narrative submission for the Schedule **shall** include the ‘Schedule Assumptions’ and ‘Data Dictionary.’ Any change, post-first submission, to the ‘Schedule Assumptions’ or ‘Data Dictionary’ requires re-submittal of the information with the affected IPMDAR deliverable.

Schedule Risk Assessment (SRA) – Reporting Frequency: An SRA can be performed, at the request of the Government, particularly when the schedule is a significant issue for the Government due to operational requirements, funding limitations, or when scheduling is considered a major risk element to the acquisition effort.

The IMS **shall** contain the contract milestones, accomplishments, criteria, discrete tasks/activities, work packages and planning packages, as applicable, from contract start to contract completion. The IMS is an integrated, logically driven, network-based schedule that is vertically and horizontally traceable. The IMS has traceability to, the project EVM WBS, project OBS, Contractor control accounts (CA), and Contractor work packages, if applicable to provide transparency for execution. The WBS and other SPD fields in the IMS are consistent with the CPD. The IMS contains all calendars that define working and non-working time periods or other information that may impact the schedule. Each event (e.g., milestone, activity, summary, etc.) identified in the IMS has a unique identifier that is consistent period to period.

All CCB authorized and funded risk burn-down (mitigation) activities **shall** be integrated into the IMS.

The Schedule consists of horizontally and vertically integrated discrete tasks/activities, consistent with all authorized work, and relationships necessary for successful contract completion. The Schedule is a single integrated network that also contains significant external interfaces, subcontractor discrete work, Government furnished equipment/information/property, and relationship dependencies for the entire contractual effort.

The Schedule **shall** contain all calendars that define working and nonworking time periods.

The Schedule **shall** reflect accurate remaining durations, start dates, and finish dates for all tasks/activities and milestones in respect to the status date.

5.2.1 If a SOW or Integrated Master Plan are used for vertical schedule integration, those references **shall** be provided in both the SPD and native schedule.

5.2.2 Milestones – Provide zero duration schedule events marking the due date for accomplishment of a specified work scope or objective. Milestone may mark the start, an interim step, or the end of one or more activities.

5.2.3 Tasks/Activities – Provide elements of work with duration and logical relationships/dependencies. Task/activity names **shall** be concise and unique in respect to other names within the Schedule. The name of each task/activity **shall** clearly reflect the scope, output

(e.g., deliverable), and place within the Schedule architecture so that the content can be understood without the subproject task structure, if applicable.

5.2.4 Duration – Provide the length of time estimated, realized, and/or remaining to accomplish a task/activity. Discrete tasks are preferred to be 42 workdays or less.

5.2.5 Baseline Dates and Information – For all tasks within the performance measurement baseline (PMB), baseline dates in the SPD and Native Schedule files **shall** be consistent with the start and finish of the cost baseline in the CPD at the control account (or work package) level. The baseline is typically set equal to the current schedule early dates at the time of the baseline establishment when change control is initiated.

5.2.6 Control Account/Work Package Identification – Tasks in the schedule must be tagged with a control account and work package as appropriate. This is accomplished by using the corresponding ControlAccountID and WorkPackageID fields in the Task table of the SPD. The SPD must be consistent with the CA and WP definitions within the CPD. As a minimum, the ControlAccountID values must be consistent with the ID tags as defined in the ControlAccount table and the Task table of the SPD must be consistent with the work package definitions.

5.2.7 Level of Effort (LOE) – LOE activities may be included or excluded in the network, as appropriate. This determination should be made based on standard procedures. LOE activities **shall** not impact discrete work or the calculation of the critical and driving paths and never drive paths. When LOE activities are included within the IMS, they are clearly identified as such and defined in the Data Dictionary.

5.2.8 Schedule Percent Complete – The calculated schedule percent complete is a time-based status calculated by the schedule tool without regard to task/activity scope accomplishment. This is not used to status budgeted cost of work performed (BCWP) (i.e., depicts the “time” percent complete based on the forecast completion date, not earned value percent complete based on work accomplished). Earned Value percent complete, when used, will be in the physical percent complete field.

5.1.2.9 Earned Value Technique – The Earned Value Technique (EVT) used for planning and statusing the Earned Value Tool can, as needed, be identified with the SPD. The EVT for each task is identified within the SPD and if LOE tasks are included within the schedule then they must be identified using the appropriate EVT tag.

5.2.10 Total Float/Slack – Provide the amount of time a task/activity or milestone forecast finish date can slip before delaying contract completion or constraint date.

5.2.11 Free Float/Free Slack – Provide the amount of time a task/activity or milestone can slip before it delays any of its successor tasks/activities or milestones.

5.2.12 Driving Path – The driving path is the longest continuous sequence chain of incomplete discrete tasks/activities/milestones and, if present, parallel chains in the schedule network that

drive the forecast dates of the agreed to discrete task/activity or milestone. A driving path may or may not be on the IMS critical path.

5.2.13 Critical Path – A sequence of discrete work packages and planning packages (or lower level tasks/activities) in the IMS that has the longest total duration with the least amount of total float/slack through an end point.

5.2.14 Subcontractor Tasks – Identify the tasks that are unique to the scope of a major subcontractor, if any. The subcontractor discrete work will be incorporated as tasks within the Contractor IMS at a level necessary for a realistic critical path.

5.2.15 Risk Mitigation Tasks – Both the Native Schedule and the SPD delivery **shall** identify items that came from the Risk/Opportunity Management System and include authorized risk mitigation activities, as applicable.

5.2.16 Schedule Visibility Tasks (SVT) – SVTs are tasks/activities or milestones in the IMS that increase management visibility and functionality of the schedule for non-PMB related items. SVTs **shall** not be used to represent any scope within the PMB. Resources cannot be assigned to SVTs, nor **shall** they be used to assess earned value performance.

5.2.17 Lead/Lag – Provide the durations of leads or lags between predecessor and successor tasks. Justification for each lead/lag **shall** be included in both the Native Schedule and the SPD.

5.2.18 Constraints – Identify the constraints applied to tasks. Justification for each constraint **shall** be included in the Native Schedule and SPD submissions.

5.2.19 Schedule Margin – A management method for accommodating schedule contingencies. It is a designated risk mitigation placeholder within the schedule and does not have assigned resources.

5.2.20 Data Dictionary for Native Schedule File – Provide a list of all Contractor defined fields, definitions, and code structures used within the Native Schedule File. The Data Dictionary **shall** be delivered with the initial submission of the Native Schedule File, and resubmitted with subsequent submissions if changes occur to the Data Dictionary, or upon request.

5.2.21 SRA – The intent of the SRA is to evaluate whether or not the baseline is executable based on the identifiable risks. An SRA is the process which uses statistical techniques to identify technical, programmatic, and schedule risks in a program and quantifies the impact of those risks on the program's schedule.

5.2.22 Custom/User-Defined Fields – Custom/user-defined fields may be required for specific information not otherwise included in the Schedule and included in the data dictionary.

5.2.22.1 Givers/Receivers – Deliverable and receivable products from one organizational element to another organization element must be identified with zero duration milestone tasks in the IMS and finish-to-start dependencies (for example, the deliverable of a subsystem, the

“giver,” to system I&T, the “receiver,” must each be a milestone task with zero-day duration linked with a finish-to-start dependency from the giver milestone to the receiver milestone). Giver and receiver milestones must be flagged or coded in the IMS.

5.2.22.2 Government Furnished Item Need Dates – Need dates for Government furnished equipment, material, property, information, facilities, or data must be identified in the IMS with milestone tasks and predecessor and successor logic.

5.2.23 Status Date – Schedule status date must reconcile with the financial and EVM calendar period end dates.

5.3 Performance Narrative Report – Detailed Analysis – Variance Analysis:

Variance Analysis Selection: Variance analysis **shall** be presented using dollars as the basis.

General: If there are no changes to the reportable element problem analysis, expected impacts, or corrective action status, the Contractor may specify “no changes since the last reported analysis” and reference the IPMDAR date when the narrative was reported.

Government Specified Variance Analysis Thresholds. Control account variances that exceed the category thresholds identified below are selected and reported:

Pool of Reportable Variances are determined by the following thresholds:

- Current cost variances exceeding +/- \$100K or +/- 10%
- Current schedule variances exceeding +/- \$100K or +/- 10%
- Cumulative cost variances exceeding +/- \$250K and +/- 10%
- Cumulative schedule variances exceeding +/- \$250K and +/- 10%
- Variances at completion exceeding +/- \$500K and +/- 10%
- Percentages determined using reporting level budgeted cost of work scheduled and BCWP as the base (denominator)

Variance thresholds **shall** be reviewed periodically to ensure the threshold levels are commensurate with the changing program risks and may be adjusted by contract modification with no change in contract price.

Performance Narrative Report – Detailed Analysis

Detailed Analysis

Required Elements: Provide explanations in the Performance Narrative Report Detailed Analysis in accordance with NASA IPMDAR Guide with the additional following definitions:

- Contract summary
- Baseline Changes: Ensure that the following are covered in the Performance Narrative Report Detailed Analysis for baseline changes:
 - Significant PMB differences (absolute values exceeding +/- 1%) as compared to the previous IPMDAR reported PMB **shall** be explained in the Performance Narrative Report Detailed Analysis.

- Significant Authorized Baseline Changes **shall** be explained in the Performance Narrative Report Detailed Analysis. “Significant Authorized Baseline Changes” are defined as the largest baseline changes occurring during the reporting period that in total represent at least 5% (dollar value) of the reporting month’s approved change requests to the PMB.
- Estimate at completion analysis
- Undistributed budget analysis
- Management reserve analysis
- Cost and schedule variance analysis
- Supplemental discussions
- Reconciliation with 533 data
- Staffing Changes: ‘Major changes’ to staffing are defined as a change of +/- 5% or greater in the reported organizational level staffing as compared to the previous IPMDAR submittal

The contract IMS **shall** be reported through delivery of the following information included in the performance narrative report but may be combined at the Contractor’s discretion with the approval of NASA’s project scheduling office. All schedule reporting **shall** be derived from, and traceable to, the Contractor’s IMS based on the Contractor’s accounting month to include the following:

- Summary Master Schedule: A one page, top level, Gantt-type schedule chart in Contractor format that summarizes the contract effort including period of performance and major milestones, critical path, schedule margin, schedule baseline, and current forecast-to-complete.
- IMS File/Database: An automated logic network file/database consisting of schedule data for all Contract Work Breakdown Structure (CWBS) elements including subcontractor effort. The entire scope of work must be decomposed into schedule tasks and milestones at a level of detail that supports discrete progress measurement and visibility into the overall design, development, fabrication, integration, assembly, test, and delivery phase of deliverables including flight hardware and software, engineering test and development units, support equipment, etc. Additionally, all schedule tasks must be integrated with dependencies that provide a total end-to-end logic network that includes a system I&T sequence at a level of detail sufficient to identify deliverable need dates. This database must contain dates for all contractual deliverables including end item delivery dates such as flight and ground system hardware and software. Government furnished equipment, material, property, information, facilities, or data need dates must be identified in the IMS. Schedule margin must be clearly defined. The database must contain a coding/flagging scheme necessary to provide sort, select, and summarize capabilities for, but not limited to, CWBS elements, project phase, level-of-effort tasks, and critical path. For contracts that comprise multiple builds, such as more than a single Instrument or spacecraft, the IMS must contain the capability to isolate the tasks by unit, serial number, vendor name, or other designation to provide the Government with visibility into baseline schedule, current forecast schedule, and schedule performance.

- Contractor Schedule Assessment Report: The Contractor Schedule Assessment Report is a narrative report accompanying the monthly IMS deliverable that **shall** contain the following:
 - Schedule Health Check Report: A status on the integrity of the Contractor's IMS, based upon an automated schedule analysis tool (e.g., NASA STAT tool, Acumen Fuse, for Project Schedule Analyzer, SSI Tools, etc.) that includes the 14 Point Assessment criteria must be submitted with rationale or corrective actions for health check results that do not meet standard "green" criteria.
 - Critical Path Report: This report must be an extract or set of views from the contract IMS and identify the critical path, secondary path, and tertiary path and include all associated tasks and milestones. The report must be submitted in a waterfall format and organized in a manner such that the path with the least amount of slack/float is delineated first, followed by each successive path according to total slack values. This report must contain a narrative explaining changes and impacts to the critical, secondary, and tertiary paths.
 - PCM Trend Report: A project control milestone (PCM) cumulative trend chart must be submitted. This report must consist of the baseline PCMs and their corresponding early finish dates, actual PCMs completed to date, and forecast PCMs for the remaining effort and their corresponding late finish dates. The PCM baseline, actual, and forecast dates must trace directly to the IMS. A listing of the PCMs that have not been completed in accordance with the schedule baseline must be included with this report.
 - Schedule Margin Trend Report: An accounting of planned and remaining schedule margin displayed in a trend format with explanations for changes during the month reported must be presented.
 - Free Slack/Float Report: A report in Contractor format must display the free slack/float for each major hardware and software deliverable to system I&T and between one subsystem/element and another. The free slack at the time the schedule baseline was set, the current reporting period's free slack, and the two prior period's free slack/float must be included.
 - Recovery and Workaround Plan Narrative: This section of the report must describe any corrective actions, recovery planning, workaround planning, or management intervention being taken or planned to hold or recover from actual and/or potential schedule delays to major contract milestones, PCMs, or givers/receivers.
- Schedule Revision Log: The Contractor must maintain and deliver a Schedule Revision Log, in Contractor format, documenting baseline changes to the IMS. At a minimum, the log should contain the following information: Baseline Change Request (BCR) number, originator, origination date, approval date, date of implementation into the IMS, BCR description and rationale for revision, CWBS affected, and impact of the revision on the overall contractual deliverable.
- Weekly I&T Schedule: Beginning one month prior to the start of system I&T activities, a weekly I&T schedule must be submitted in Contractor format. The I&T schedule must be delivered on Friday of each week beginning with the start of I&T through completion of the Contractor's I&T effort. The I&T schedule must be utilized as the I&T manager's daily working schedule and must contain details of the activities accomplished in the

previous week (with the exception of the first delivery) and details of work planned for the next two weeks. The I&T schedule must be organized by shift when appropriate (if more than one standard shift). The weekly I&T schedule deliverable is separate from, and in addition to, the monthly IMS deliverable. The weekly I&T schedule must be submitted in electronic format (Adobe PDF is acceptable).

- SRA: Upon request by the Government, the Contractor must provide the results of its probabilistic SRA, including the probability of meeting contractual deliveries and milestones in histogram/S curve formats. The SRA results must also include rank order “tornado diagrams” of both the risks and tasks affecting the completion of contractual milestones and deliverables.

DATA ITEM DESCRIPTION

1. CDRL No.:

PM-4

2. Title:

INTEGRATED MASTER SCHEDULE (IMS)

3. Reference:

400-PG-7120 Schedule Management

4. Use:

To track the design, build, and I&T status of all items under contract over contract lifetime with a coherent, integrated scheduling tool. The IMS will be used to plan, monitor, communicate issues and control all activities, including pertinent resources and facilities necessary to accomplish assigned tasks in compliance with the SOW. Schedule will be used to provide management insight into Contractor status, potential problem areas and critical path identification which will serve as the basis for evaluating Contractor performance. The baseline IMS will be the basis for evaluating the impact of Government-directed changes to the Launch date.

5. Preparation Information:

The IMS **shall** be developed and maintained in the release of Microsoft Project at contract award using the Critical Path Method scheduling technique. It will consist of the schedule baseline, actual completion status, percent complete, and the current schedule forecasts that is updated for each reporting period. Reference 400-PG-7120, Schedule Management, for best practices to consider in developing and managing the IMS.

The IMS **shall** contain the contract milestones, accomplishments, criteria, discrete tasks/activities from contract start to contract completion (or the awarded CLINs/options).

The IMS contains all calendars that define working and non-working time periods or other information that may impact the schedule. Each event (e.g., milestone, activity, summary, etc.) identified in the IMS **shall** have a unique identifier that is consistent period to period.

All CCB authorized and funded risk mitigation activities **shall** be integrated into the IMS.

The IMS will provide the Contractor's time-phased plan, current status, key milestones, schedule reserve, task interdependencies, and major development phases necessary to accomplish the total scope of work.

The Contractor **shall** provide a schedule that reflects adequate reserve margins for the following:

- From contract award through Instrument delivery for awarded CLINs/Options.
- From delivery to the launch date in the contract.

2.2 Review Data Items

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-1

2. Title:

KICKOFF MEETING DATA PACKAGE

3. Reference:

SOW 1.5.1.1

4. Use:

To provide team introductions and an overview of the effort's management and technical approaches, the design concept, concept of operations, development plans, and Safety and Mission Assurance (SMA) processes.

During the Production Phase, the Kickoff Meeting is used to identify any differences or updates to design or processes from the flight models produced under previous CLINs. If there are minimal changes, the Kickoff for this CLIN may be combined with the Manufacturing Readiness Review (MRR).

5. Preparation Information:

The Kickoff Meeting agenda will be developed jointly by the Contractor and Government.

The Kickoff Meeting Data Package should include overviews and summaries of the following:

- Contractor's and Government project's organization and personnel
- Contractor processes to be applied to this contract such as risk management, engineering, analysis, configuration management, SMA, quality, verification, material, parts, and failure review processes
- OBS
- IMS summary with detail for the near-term schedule (between kickoff and IBR)
- Planned periodic meetings and a notional meeting schedule for the first 6 months (PMRs, table top reviews, peer reviews, program milestone reviews, etc.)
- Instrument design concept
- Technology development work, status and plans to mature
- Engineering Development Program plans
- Current Risk List
- Contractor facilities (tour if possible)

- Identification of key suppliers/subcontractors and overview of long lead item procurement plans (e.g., what are the long leads, planned start, finish, need)
- Identification of any areas of requirements uncertainty for the team to work together to resolve, possible splinter discussion.

Production CLIN Kickoff Meeting Data Packages should include:

- Modifications from previous unit – changes to design, process (including Mission Assurance), qualification approach, etc.
- Rationale for modification
- Implications resulting from the modification
- Test approach changes resulting from the modification
- CDRL/Deliverable updates as a result of the modification

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-2

2. Title:

SYSTEM REQUIREMENTS REVIEW
(SRR)/SYSTEM DEFINITION REVIEW (SDR)
DATA PACKAGE

3. Reference:

GSFC-STD-1001 Section 4.2

4. Use:

To evaluate the requirements, requirements flow-down, and the operational concepts and to validate the realism of the functional and performance requirements and their congruence with the system configuration selected to conduct the mission.

5. Preparation Information:

- The SRR/SDR Data Package **shall** contain:
- All relevant Instrument information required to satisfy paragraph 4.2 (Criteria for Successful Completion) of GSFC-STD-1001, Criteria for Project Flight Critical Milestone Reviews.
- Specification tree illustrating the Contractor's specifications starting at the top-level Instrument Performance Specification, showing all lower-level specifications to the sub-assembly (box/board) level, and indicating the relationships and interfaces between the documents.
- Identification of driving requirements.
- Identification and description of all single point failures (SPF).
- Overview of the development approach (simulations, modeling, reviews, engineering models, bread boarding, testing, etc.)
- Overview of critical GSE/special test equipment (STE), long leads expected, and higher risk areas of the design effort.
- Minutes that consist of a list of meeting attendees, copies of presentation material provided at the review not in the presentation charts package, action items assigned, and agreements made.

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-3

2. Title:PRELIMINARY DESIGN REVIEW (PDR) DATA
PACKAGE OR DELTA PDR3. Reference:

GSFC-STD-1001 Section 5.2

4. Use:

To demonstrate the Flight Equipment and GSE/STE design meet the documented requirements.

5. Preparation Information:The PDR Data Package **shall** contain:

- All relevant Instrument information required to satisfy paragraphs 5.2 (Criteria for Successful Completion) of GSFC-STD-1001, Criteria for Project Flight Critical Milestone Reviews.
- Information to cover the Instrument data processing algorithm design.
- Design and significant scope changes since the last review.
- Instrument error budgets and performance budgets including Beginning of Life to End of Life performance degradation estimates.
- Interface designs and analyses, including the following information:
 - Preliminary analyses that show that the "swept" or deployed volume meets the volume requirements defined in the Performance Requirements Document (PRD), ICD, and MICD, accounting for all distortions and misalignments.
 - Preliminary estimates of masses and inertias to permit sizing the spacecraft control components to meet pointing and stability requirements.
 - Preliminary analyses of disturbances of the sum of any periodic disturbance torques, in order to produce the corresponding magnitude spectrum.
- GSE design and workflow, and describe how each item will be fabricated, tested, and certified when needed.
- Manufacturing, assembly, integration, test flow, and calibration/test plans.
- Preliminary Failure Modes and Effects Criticality Analyses (FMECA) and results.
- SPFs.
- List of long lead items, and of items that may become obsolete prior to completion of all flight instruments, identify those items that must be procured prior to CDR (including a list of those that were ordered prior to PDR and SRR), and the plan for procuring these items and all parts.

- Contamination control requirements and processes to be applied.
- Status of each document required at PDR and their readiness for use.
- All Instrument risks and their mitigation approach.
- Status of all sub-contracts and the preliminary design status of subcontracted critical assemblies and sub-assemblies.
- Status of the Engineering Development Program.
- Summary of all breadboard and brassboard testing the available test results.
- Development status of sub-assembly engineering units, and available test data.
- Significant subcontract/procurement status and approach to SMA requirements.
- Software development status, process, and software assurance process.
- Specification tree illustrating the Contractor's specifications starting at the top-level Instrument Performance Specification, showing all lower-level specifications to the sub-assembly (box/board) level, and indicating the relationships and interfaces between the documents.
- Minutes of design reviews that **shall** consist of a list of meeting attendees, copies of presentation material provided at the review not in the presentation charts package, action items assigned, and agreements made.

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-4

2. Title:CRITICAL DESIGN REVIEW (CDR) DATA
PACKAGE OR DELTA CDR3. Reference:

GSFC-STD-1001 Section 6.2

4. Use:

To present the Instrument Flight Equipment and GSE/STE design and operation, and to demonstrate that all related manufacturing documentation, processes, and fixtures are in place before hardware manufacture of new or improved hardware begins, and to demonstrate that the design meets all performance requirements.

5. Preparation Information:

The CDR Data Package **shall** contain:

- All relevant Instrument information required to satisfy paragraphs 6.2 (Criteria for Successful Completion) of GSFC-STD-1001, Criteria for Project Flight Critical Milestone Reviews.
- Instrument error budget and performance budgets including Beginning of Life to End of Life performance degradation estimates.
- Procurement status of long lead items and Electrical, Electronic, and Electromechanical (EEE) parts.
- Manufacturing flow, and the status of manufacturing and assembly drawings, bill of materials, etc.
- Status of manufacturing procedures and an overview of the manufacturing approaches.
- Special or unique SMBA fixtures and test equipment.
- Overview of the facilities required for manufacturing.
- Detailed analysis and results from the FMECA.
- Worst case analyses of:
 - Electrical circuits
 - Scanning drive system
 - Lubrication and lubrication loss
 - Tolerance and tolerance sensitivity analyses (including thermal and mechanical considerations)
- Stress analyses
 - Use of NASTRAN is encouraged

- Hand calculated verifications of a few parameters to check the model is recommended (e.g., overall mass, reaction forces, rough estimates, etc.)
- Thermal analyses of all subsystems.
- Test plans (including all environmental tests).
- Maintainability considerations, including storage.
- Materials and processes lists.
- Summary of deviations/waivers.
- Contamination control and monitoring considerations including contamination allowances and budgets and methods to ensure necessary contamination mitigation.
- Spares program quantities, plans, schedules, status.
- System safety hazards analyses:
 - Identification of hazards (e.g., hazards identification matrix)
 - SPF summaries
 - Risk assessment rationale
- Status of incomplete deliverable documentation due at CDR and readiness for use.
- Status of the Engineering Development Program
- Test data from sub-assembly engineering development units.
- Development status of all GSE/STE, including test and calibration procedures, and the software/firmware design and operation and interface aspects as evaluated since the PDR.
- Status of all program risks and their mitigation.
- Qualification history/status and details for each component for the flight instruments including data.
- Specification tree illustrating the Contractor's specifications starting at the top-level Instrument Performance Specification, showing all lower-level specifications to the sub-assembly (box/board) level, and indicating the relationships and interfaces between the documents.
- Minutes of design reviews that includes of a list of meeting attendees, copies of presentation material provided at the review not in the presentation charts package, action items assigned, and agreements made.

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-5

2. Title:PRE-ENVIRONMENTAL REVIEW (PER) DATA
PACKAGE OR DELTA PER3. Reference:

GSFC-STD-1001 Section 9.2

4. Use:

To present the description and results of the Instrument Pre-Environmental Test program, and demonstrate readiness for environmental testing.

5. Preparation Information:

The PER Data Package **shall** contain:

- All relevant Instrument information required to satisfy paragraph 9.2 (Criteria for Successful Completion) of GSFC-STD-1001, Criteria for Project Flight Critical Milestone Reviews.
- Analyses and reports required at the review and any updates since CDR.
- Failure report summaries including status of action and rationale for closure.
- Results of the functional and interface tests.
- Descriptions of any malfunctions and their status and the corrective actions.
- Comparison of measured performance with requirements and discussion of the effect of any variance and waivers.
- I&T and mission operation constraints.
- Contamination avoidance processes and approaches, approaches for facility contamination control, packaging and protection, and contamination mitigation during test operations.
- Approach to meeting safety requirements.
- List of spares for flight equipment and GSE/STE including available quantities, usage to date, plans for replenishment, schedule for remaining deliveries.
- Review of flight hardware handling procedures.
- Spacecraft interface concerns, problems, and solutions, including:
 - Analysis and test data that demonstrates that the "swept" or deployed volume is verified, accounting for all distortions and misalignments.
 - In electronic format, time/magnitude plots of disturbances, for decomposition, of the sum of the Instrument-induced periodic disturbance torques, in order to produce the corresponding magnitude spectrum.

- Orbital operations plans and status of documentation and databases.
- Compatibility of Instrument with spacecraft flight support equipment, GSE and operational ground equipment.
- Status of all program risks and their mitigation plans.
- Minutes of design reviews includes a list of meeting attendees, copies of presentation material provided at the review not in the presentation charts package, action items assigned, and agreements made.

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-6

2. Title:

PRE-SHIP REVIEW (PSR) DATA PACKAGE

3. Reference:

GSFC-STD-1001 Section 11.2

4. Use:

To evaluate Instrument performance during qualification or acceptance testing and evaluate readiness to ship.

5. Preparation Information:

This data package **shall** address, as a minimum:

- All relevant Instrument information required to satisfy paragraph 11.2 (Criteria for Successful Completion) of GSFC-STD-1001, Criteria for Project Flight Critical Milestone Reviews.
- Solutions to all problems encountered during the environmental test and validation program and the solution rationale.
- Any rework/replacement of hardware, regression testing, and test plan changes.
- Compliance with the Test Verification Matrix (CDRL IT-1) and SPVM (CDRL SE-1).
- Measured test margins versus requirements.
- Instrument performance budgets including Beginning of Life to End of Life performance degradation estimates.
- Qualification/acceptance temperature margins.
- Any data that has been trended to identify compliance with specification, indicating a change or drift to the trend.
- Summary of the analyzed monthly trend reports (CDRL SE-2) of trended Instrument data, focusing on trends that have changed or drifted anomalously.
- Total failure-free operating time of the item.
- Number of cycles during testing of parts with finite lifespans.
- Results of the final audit of any remaining drawing changes.
- Update from CDR on shipping containers, monitoring, transportation, and control plans and procedures.
- GSE status.
- Post shipment plans.
- Spacecraft integration plan.
- Plans for storage of the Instrument, if required.

- Documentation verifying that the Instrument meets contamination budgets and requirements.
- Alignment data relative to the Instrument Boresight and the mounting surface datum(s) associated with the Instrument.
- Minutes of design reviews that includes a list of meeting attendees, copies of presentation material provided at the review not in the presentation charts package, action items assigned, and agreements made.

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-7

2. Title:

INTEGRATED BASELINE REVIEW (IBR)

3. Reference:

4. Use:

To assess and validate the Contractor's PMB by conducting an Integrated Baseline Review (IBR).

5. Preparation Information:

The Contractor **shall** conduct a formal on-site IBR. This review will be conducted as early as practicable and conducted within 180 days of contract award or formal contract definitization. Additional baseline reviews may be required when substantial changes are made to the baseline (e.g., subsequent detailed planning, including routine rolling wave planning and program re-planning resulting from incorporation of significant contract modifications). The IBR will be tailored to meet the needs of the SMBA project.

The Government will coordinate the timeframe and scope of the IBR with the Contractor. The IBR will ensure that the PMB captures the entire scope of work, is consistent with the contract schedule requirements and has an adequate resource plan.

The IBR team **shall** include Government personnel.

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-8

2. Title:ENGINEERING PEER REVIEW DATA
PACKAGES

3. Reference:

4. Use:

EPRs focus on the design and implementation details at levels that system-level reviews cannot address. They provide a resource for design teams to identify potential engineering design and implementation flaws, and increase the probability of success. Applying the EPR process early and throughout the product life cycle affords the maximum advantage in terms of resource efficiency as well as design confirmation and ultimate mission success. Peer review documentation represents knowledge that may prove invaluable later.

5. Preparation Information:

EPR documentation **shall** be in Contractor format and comply with the submitted EPR plan (CDRL PM-2).

The Contractor **shall** prepare minutes of design reviews that includes a list of meeting attendees, copies of presentation material provided at the review not in the presentation charts package, action items assigned, and agreements made.

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-9

2. Title:MANUFACTURING READINESS REVIEW
PACKAGE3. Reference:

4. Use:

The purpose of the MRR is to verify the engineering baseline by a design, drawing, and engineering requirement review to assure that the release to manufacturing represents the final and correct flight equipment and that the manufacturing planning is complete and reflects the SMBA Instrument configuration.

5. Preparation Information:

The content of this review package **shall** be as follows:

- Process flowchart
- Schedule
- Product description
- Configuration Item Identification List (CIIL)
- Computer Software Configuration Item (CSCI)
- Summary of any changes to configuration baseline, including, parts, materials, and processes
- Summary of any updates to the CDR data submittals, design analysis, and SMBA plans and procedures provided under the NASA SMBA Contract
- Detailed manufacturing schedules
- Manufacturing flow diagram including inspection, mandatory inspection points, and test points
- Drawing readiness
- Process readiness
 - List of drawings
 - Procedures
 - Machine availability and certification
 - Process qualification
 - Training and personnel qualification
 - Workmanship standards
- Product FMECA (manufacturing risk mitigation)
- Special process list

- ESD control plan and cleanliness control
- Material availability
- Personnel availability
- Tooling readiness
- Facility readiness
- Parts status and qualification status on piece parts
- Summary of all nonconformance and test failure reports initiated for the prior flight unit including the status of all corrective/preventive action.
- Status of suppliers
- Status of prohibited material screening
- A table listing all in-process and end item procedures where CTP, CTB, and/or Consent to Integrate (CTI) decision points require Government approval
- Minutes of the review that includes a list of meeting attendees, copies of presentation material provided at the review not in the presentation charts package, action items assigned, and agreements made.

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-10

2. Title:

TEST READINESS REVIEW (TRR)

3. Reference:

GSFC-STD-1001

4. Use:

A TRR ensures that the test article (hardware/software), test facility, support personnel, and test procedures are ready for testing and data acquisition, reduction, and control. A TRR is held prior to commencement of verification or validation testing. The objectives of the TRR are to:

- Ensure a thorough review of the products supporting the review.
- Ensure the products meet the entrance criteria and success criteria.
- Ensure issues raised during the review are appropriately documented and a plan for resolution is prepared.

A successful TRR signifies readiness for formal test initiation.

5. Preparation Information:

At the TRR and within the TRR data package, the Contractor **shall** show that:

- Adequate test plans are completed and approved for the system under test.
- Adequate identification and coordination of required test resources are complete.
- Previous component, subsystem, and system test results form a satisfactory basis for proceeding into planned tests.
- All known system discrepancies have been identified and disposed in accordance with an agreed-upon plan.
- Risks associated with the test have been identified and accepted by program/competency leadership as required.
- Plans to capture any lessons learned from the test program are documented.
- The objectives of the testing have been clearly defined and documented, and the review of all the test plans, as well as the procedures, environment, and the configuration of the test item, provide a reasonable expectation that the objectives will be met.
- The test cases have been reviewed and analyzed for expected results, and the results are consistent with the test plans and objectives; test personnel have received appropriate training in test operation and safety procedures.
- Roles and responsibilities of all test participants are defined and agreed to.

- All required test resources people (including a designated test director), facilities, test articles, test instrumentation, and other enabling products are available to support required tests.
- Test contingency planning has been accomplished.

The Contractor **shall** prepare minutes of the review that includes a list of meeting attendees, copies of presentation material provided at the review not in the presentation charts package, action items assigned, and agreements made.

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-11

2. Title:INPUTS TO SATELLITE/PROJECT LEVEL
REVIEWS3. Reference:

4. Use:Inputs to satellite/project level reviews.

5. Preparation Information:

The Contractor **shall** support satellite/project level reviews by contributing to, preparing, reviewing, and/or presenting presentations and documentation such as Instrument overview or design summary, plans, procedures, test, spacecraft interfaces and data evaluation summaries, etc.

2.3 Systems Engineering Data Items

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-1

2. Title:

SYSTEM PERFORMANCE VERIFICATION
PLAN (SPVP) AND MATRIX (SPVM)

3. Reference:

4. Use:

The System Performance Verification Plan (SPVP) provides the overall approach for accomplishing the verification program. Defines the specific tests, analyses, calibrations, alignments, hardware models, etc., that will demonstrate that the flight hardware complies with the mission requirements. The SPVP will include the System Performance Verification Matrix (SPVM).

5. Preparation Information:

The SPVP **shall**:

- Provide an overview of the entire Instrument verification program.
- Describe the verification approach to the SMBA Instrument PRD.
- Describe the SMBA PRD and SOW requirement flow down approach to:
 - SMBA Instrument Performance Specification (CDRL SE-5)
 - Inputs to the spacecraft to SMBA ICD and MICD (CDRL SE-11)
 - Instrument Integration, Test, and Verification Plan (CDRL IT-1)
 - Calibration/Validation Plan (CDRL CV-1)
 - Instrument On-Orbit Commissioning Plan (CDRL OO-1)
- Describe the dependence if verification relies on measurements, tests, or analyses at lower (or other) levels of assembly.
- Document plans to verify by test or inspection the proper polarity, orientation, and position of all components (sensors, switches, and mechanisms).
- Include level of assembly, configuration of item, objectives, test phases, appropriate functional operations, and requirements for procedures and reports.
- Describe which level of assembly was chosen for verification of any PRD requirement not verified at the Instrument level with rationale.
- Include analysis objectives, a description of the mathematical model, assumptions on which the model will be based, required output, criteria for assessing the acceptability of the results, interaction with related test activity, and requirements for reports.

-
- Be cross referenced to the Program Risk Register any PRD requirement identified in the Instrument Integration, Test, and Verification (CDRL IT-1) as having test or demonstration limitations and/or the On-Orbit Commissioning Plan (CDRL OO-1) as having any on-orbit verification tests.
 - Provide a verification burn-down plan for the SMBA PRD requirements.

The SPVM **shall** stipulate how each Instrument requirement in the Instrument PRD will be met for each flight unit.

For each SMBA PRD requirement, the SPVM **shall** include at a minimum in Excel format the following content:

- Paragraph number and heading
- PRD Requirement Reference ID (e.g., SMBA-1446)
- Short requirement summary description
- Full PRD Requirement Text
- PRD requirement applicable flow-down document(s) and its section or requirement ID
- Verification Level (e.g. Antenna Subsystem, Instrument, satellite)
- Environment in which the PRD requirement is verified
- Applicable test procedures
- Verification Unit applicability by flight model (e.g. FM1, FM2, All, etc.)
- Verification applicability by Instrument serial number
- Verification Method (e.g., test, analysis, etc.)
- Expected compliance/non-compliance with summary of the expected measured/analyzed/demonstrated performance
- Verification Trigger Event(s) that collects the artifact(s)
- Expected Verification Phase that the Government approves the verification in Dynamic Object-Oriented Requirements System (DOORS) (e.g., PDR, CDR, PER, etc.)
 - o If non-compliance, describe non-compliance impact, reach, and mitigation
- Artifact Descriptive Name (e.g., TVAC Test Report, EMI Analysis, etc.)

A DOORS compatible version of the SPVM **shall** be provided.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-2

2. Title:TREND AND TPM PLAN/ANALYSIS,
REPORTS, AND OPERATIONS LOG3. Reference:

4. Use:

Provides a list and a means of tracking critical engineering and performance parameters for the Instrument and across Instruments. Provides a log of Instrument operating hours. A brief rationale for including the parameter **shall** be included. The list **shall** be coordinated with the Government prior to implementation. The trend list **shall** be reviewed with the on-orbit operational trend parameters prior to launch.

5. Preparation Information:

The trend plan **shall** include:

- The list of Technical Performance Metrics (TPM) to be monitored for degradation trends in Instrument performance, reliability, and out of family with other Instruments.
 - Particular attention should be made to track worse case Instrument science performance in the areas of digital counts, gain and offset, noise equivalent differential temperature (NEdT), noise power spectral densities, calibration uncertainty, 1/f noise, inter-channel noise correlation, and residual nonlinearity.
- Collecting and trending parameters starting at component acceptance testing through system I&T test phases through on-orbit commissioning.
- The trend list is to be reviewed with the on-orbit operational trend parameters prior to launch.
- The trends parameter list to include limit information and descriptions of the hazards indicated in sufficient detail to allow the Mission Elements Provider to respond appropriately to out-of-limit parameters.
- An operations log is to be maintained of the accumulated operating time of the Instrument.

The trend plan, trend reports, and operations log **shall** be delivered in Contractor's format.

Suggested TPMs for trending Pre-Launch:

- Mass (kg)
- Operational Peak Power (W)
- Operational Average Power (W)

- Survival Peak Power (W)
- Average Data Rate (Mbps)
- Peak Data Rate (Mbps)
- Memory Utilization
- CPU Utilization
- Gain Stability
- Center Frequency Stability
- Bandwidth
- Out-of-band Rejection
- Passband Ripple
- Pointing Accuracy
- Beamwidth
- Beam Efficiency
- Counts (channel, warm load, cold view)
- Gain and Offset
- NEdT
- Noise Power Spectral Densities
- Calibration Uncertainty
- 1/f Noise Ratio
- Inter-channel Noise Correlation
- Residual Non-linearity

Suggested TPMs for continued trending Post-Launch:

- Counts (channel, warm load, cold view)
- Gain and Offset
- NEdT
- Noise Power Spectral Densities
- Calibration Uncertainty
- 1/f Noise Ratio
- Inter-channel Noise Correlation
- Residual Nonlinearity

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-3

2. Title:

ENGINEERING DRAWINGS

3. Reference:

4. Use:

To provide layouts and engineering drawings to serve as the basis for technical discussions, evaluations, manufacturing, fabrication, assembly, test, operations, and maintenance. To satisfy some of the satellite drawings requirements of the launch services provider.

5. Preparation Information:The Contractor **shall**:

- Maintain an indentured drawing list of all drawings (including drawings from subcontractors).
- Make that list available virtually at Top-Level Instrument Reviews.

The Contractor should show quick-reference relationships between drawings and the next level of assembly.

All engineering drawings **shall** be in the Contractor's designated format.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-4

2. Title:ENGINEERING CHANGE REQUESTS,
DEVIATIONS, AND WAIVERS

3. Reference:

4. Use:

To facilitate the orderly processing of engineering change requests to specifications or requirements, specification deviations and waivers, to appropriate level of review, classification, and approval authority for disposition.

5. Preparation Information:

Engineering Change Requests (ECR) **shall** be created for all proposed Class I changes.

Requests for Waivers and Deviations **shall** include:

- Description of the requirements that apply to the item, subsystem, or system needing the waiver or deviation, and provide specific information regarding the nature of the noncompliance.
- Description of the effect of the proposed noncompliance on item, subsystem, and system performance at higher levels of assembly.
- Description of the cost or schedule impact of non-approval of the waiver or deviation.

For the purposes of this DID, a Class I ECR is a change that:

- Affects any Government Contract requirement, specification, or interface requirement.
- Affects schedules of end item deliverables to the Government.
- Impacts Government Furnished Equipment.
- Affects configuration to the extent that changes would be required to prior deliverables in order to maintain specified performance.
- Creates a new SPF possibility.

A change may be classified Class II when it does not fall within the definition of a Class I change as given above. Drawing changes that do not affect a baseline, interface, etc.

Class II changes normally do not require Government CCB approval unless they are written against Government CM-controlled documents. Examples of Class II changes are:

- A change in documentation only (e.g., correction of errors, addition or clarifying notes or views).

- A minor change in hardware which does not affect any item listed under Class I changes.
- Drawing changes that do not affect a baseline, interface, etc.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-5

2. Title:INSTRUMENT PERFORMANCE
SPECIFICATION

3. Reference:

4. Use:

To document the Instrument system and subsystem specification derived from the Instrument PRD and the other contract requirements documents to ensure that the Instrument design requirements flow correctly from the Instrument PRD and the other contract requirements documents to lower level requirements.

The Instrument Performance Specification **shall** indicate the flow-down and allocation of all SMBA technical (functional and performance) requirements to lower levels (including subsystems, software, etc.).

5. Preparation Information:

The SMBA Instrument Performance Specification **shall:**

- Include Beginning of Life (BOL) performance requirements derived from the PRD End of Life (EOL) requirements.
- Be delivered in the Contractor specified format.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-6

2. Title:STRUCTURAL AND MECHANICAL
ANALYSIS AND MODELS

3. Reference:

4. Use:

To document the analyses and data which demonstrates that all structural and mechanical subsystem performance requirements are satisfied. The analysis will be used to determine the Instrument structural adequacy to withstand transportation, launch, and on-orbit loads. Also, they will show compliance with frequency requirements. The satellite/launch vehicle loads analysis will be used in the development and definition of the Instrument/spacecraft interface loads. The model will be utilized by the spacecraft prime contractor for system-level modeling.

5. Preparation Information:

The structural analysis report(s) **shall** include, but **shall** not be limited to:

- The stress analysis of structural elements to design limit loads and environments.
- Analysis on each separately-mounted Instrument component.
- Deflection analysis.
- Decompression/venting analysis.
- Dynamic response of the printed wiring boards and chassis walls containing subassemblies.
- Test results and compliance with load and environment requirements.

The structural analysis report(s) should contain:

- All assumptions and design criteria used in the analysis, descriptions of the analytical models, and results, conclusions, and recommendations.
- All detail calculations necessary to substantiate results; computer detailed calculations other than standard outputs are exempt.

Structural and Dynamic Mathematical Models **shall** be verified with dynamic test data and models updated to agree with the structural test results.

The Contractor **shall** develop and document test-verified mathematical structural models of the SMBA in the launch and on-orbit configuration. These **shall** include:

- A structural finite element model and a dynamic model in Craig-Bampton form.

- The finite element model **shall** represent the structural and dynamic characteristics of the Instrument.
- Interface degrees of freedom **shall** be compatible with corresponding attachment degrees of freedom on the model of the spacecraft.
- The dynamics model **shall** be based on the finite element structural model using standard reduction techniques.

The structural finite element models and dynamic models **shall** be:

- Provided on electronic medium in a format acceptable to the project.
- Accompanied by documentation necessary for use of the model.
- Delivered for use in the verification loads cycle analyses.

The structural finite element models and dynamic models should be in NASTRAN format for commonality across the Instrument, spacecraft, and launch vehicle.

The finite element model **shall** include the following:

- A listing of the input data for the model.
- Model definition plots, coordinate system definition, definition of stiffness properties, nodal mass distribution, and any other pertinent model definition information, along with documentation of the correlation between the modeled properties and the Instrument design.
- Mode shapes, frequencies, modal damping, modal participation factors, modal effective weights, orthogonality and cross orthogonality between analytical and test modes, and all data required to demonstrate test-verification of the models, and
- Characterization of all significant frequencies and mode shapes for the Instrument constrained at the mechanical interface to the spacecraft, in addition to the constraint modes, one for each boundary degree of freedom.

The dynamic models **shall**:

- Include the entire Instrument and any auxiliary equipment.
- Be in Craig-Bampton form with modes that represent the dynamic characteristics of the Instrument to at least 200 Hertz (Hz), with all modes through 50 Hz test verified by dynamic modal survey testing.
- Define dynamic degrees of freedom to allow calculation of acceleration levels and relative deflections at critical points.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-7

2. Title:

THERMAL ANALYSIS MODELS

3. Reference:

4. Use:

To document the detailed and reduced thermal analyses used in the design of the Instrument. Several analysis reports and the Instrument thermal models themselves are required to be delivered as noted in the table of CDRL deliverable models and reports.

The Instrument reduced thermal model will be integrated with the satellite thermal model. The satellite thermal model will be utilized by the spacecraft prime contractor to characterize satellite thermal performance. The satellite thermal model will be used to calculate radiation exchange factors, environmental heating rates (solar, albedo, Earth infrared) during various phases of the mission including launch, ascent, on-orbit operations, all survival modes, and including Instrument calibration maneuvers. The satellite thermal model will be used to characterize the thermal response of the satellite for all spacecraft states and Instrument modes over the entire length of the mission. The satellite thermal model will also be used to simulate the thermal environments encountered during the satellite environmental test campaign.

5. Preparation Information:

The analyses **shall** contain, as a minimum, the following:

- All pertinent assumptions used for analysis, description of the thermal model with associated boundary conditions.
- All detailed calculations needed to substantiate the results (if used).
- The temperature and mechanical effects on the unit in thermal vacuum (if required) versus ambient pressure, and results, conclusions, and recommendations.
- References/measurements cited to support assumptions about interface conductances and optical and thermophysical properties.

The detailed thermal analysis **shall** demonstrate that all internal parts meet established thermal design criteria, based upon program de-rating or electrical performance requirements.

The associated reports **shall** describe in detail the various analysis cases (e.g., launch, cold survival, worst case hot and cold operational, nominal science, etc.). This description **shall** include:

- List of all parameters and assumptions.

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- Relationships between various model, or sub-models, and how boundary conditions are obtained.
 - Detailed calculations and descriptions of computer models and analyses.
 - Results of the final analyses including predicted thermal performance for worst case extremes and design margins.

Instrument Reduced Thermal Model Requirements:

The Instrument Contractor **shall** provide a reduced thermal model using ThermalDesktop (geometry) and SINDA/FLUINT (math model).

The thermal model **shall** include:

- An adequate level of detail to predict, under worst-case hot and cold environments:
 - All critical temperatures, including those that drive operational and survival temperature limits and heater power.
- Worst-case conditions that include variations in season, orbit selection, orbital time, and environmental flux parameters (seasonal and spatial) and a rational combination of the effects of design tolerances, fabrication uncertainties, material differences, and degradation due to aging.
- Necessary detail to support thermal distortion studies, pointing budget assessments, and/or power budget activities as required.
- The geometry used to represent Instrument radiators, apertures, optical elements, mounting feet, and any other components identified as ‘critical’ in the thermal model **will** be congruent with the geometry found in the mechanical/CAD model.
- Instrument thermal models may deviate from their mechanical counterparts to account for soft goods, such as multi-layer insulation blankets, or to intentionally exclude unnecessary features.
- A user’s guide that provide a complete description of the model and any additional information required for model execution.

Instrument Thermal Analysis report **shall** describe in detail the various analysis cases (e.g., launch, transfer orbit, on-orbit, on orbit hot and cold cases, etc.).

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-8

2. Title:

CONTAMINATION ANALYSIS REPORT

3. Reference:

4. Use:

To enable the Government to make informed decisions about requirements allocation and deviation or waiver approvals by providing insight into the Contractor's contamination requirements and margins.

5. Preparation Information:

The Contractor **shall** deliver an Instrument internal mass transport analysis based on Instrument geometry, internal component locations, and the materials used to build the hardware. Materials selection and location is a key factor in determining where contaminants will move. The mass transport analysis results **shall** include at a minimum:

- Description of math model used, tabulation assumptions, critical parameters.
- Reference the mechanical model version and thermal model version used to generate finite element model and the temperature predicts, respectively.
- List the deposition onto sensitive surfaces (usually cold surfaces) for the duration of mission life as well as the temperature history of these elements.
- List the surfaces of concern as contamination sources.
- Provide a thorough understanding of the outgassing paths within the Instrument can help understand contamination aspects of potential flight problems that may be contamination related.
- Outgassing rates of the Instrument and its internal hardware based on temperature predicts for typical operating, hot and cold cases for each node.
- Determine the necessary levels of bakeout required to insure proper performance of the Instrument.
- The maximum amount of contaminants allowed onto sensitive surfaces and apertures for the duration of mission life in order to satisfy throughput requirements for optics.

The Contractor **shall** deliver the chamber bakeout performance analysis. Chamber bakeout performance analysis results **shall** include:

- Chamber pumping efficiency
- Layout and operating parameters of vacuum chamber

- Detailed layout of the arrangement of the hardware within the chamber
- Temperatures of all chamber components, and locations of all vents and cold traps
- Location and operating temperature of the Thermal Quartz Crystal Microbalances (TQCM) used to monitor the bakeout/certification

Contamination analysis report **shall** correlate model results with Instrument-level thermal vacuum test outgassing results and verify PRD requirements.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-9

2. Title:

EMI/EMC ANALYSIS REPORT

3. Reference:

PRD 5.7.5, 6.2, 7.3

4. Use:

To document the Electromagnetic Interference (EMI)/Electromagnetic Compatibility (EMC) analyses used in the design of the Instrument.

5. Preparation Information:

The EMI/EMC analysis at a minimum **shall** address:

- EMI/EMC design features of the Instrument and any design changes from previous Instrument builds.
- Instrument shielding design analysis for both radiation emissions and radiated susceptibility.
- Instrument radiated susceptibility both direct and indirect coupling.
- Compatibility between an Instrument and all internal sources and receivers on the satellite.
- Compatibility between an Instrument and all external sources and receivers "visible" on the ground, on the launch vehicle, and on-orbit.
- Instrument Survival Analysis
 - ON condition
 - OFF condition for factory, transport, launch pad, and ascent
- The verification of design requirements in EMI/EMC Sections of PRD.

To be provided in Contractor preferred electronic format.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-10

2. Title:INSTRUMENT CONCEPT OF OPERATIONS
DOCUMENT

3. Reference:

4. Use:

To provide information on the Instrument Concept of Operations derived from unique aspects of the Contractor's design.

5. Preparation Information:

The Instrument Concept of Operations Document **shall** describe the methodology required to operate the Instrument. Relative to the Contractor's design (including any and all Instrument unique operational features), the operations concepts **shall** address:

- Instrument operational environment and limitations
- Operational requirements over and above existing Instrument requirements necessary to accommodate design unique factors
- Unique ground command and data handling operations
- User support operations
- Contingency and emergency Instrument unique operational scenarios, including Flight Software (FSW) changes
- Radio Frequency Interference (RFI) detection related activities
- Hyperspectral configuration related activities.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-11

2. Title:INPUTS TO THE SPACECRAFT TO
INSTRUMENT INTERFACE CONTROL
DOCUMENT INCLUDING MECHANICAL ICD3. Reference:

4. Use:

To establish interface definition and control between SMBA and the NEON spacecraft.

5. Preparation Information:

This document **shall** provide detailed technical information regarding the interface of the SMBA to the spacecraft. The inputs to the spacecraft to Instrument ICD **shall** include interface requirements for the SMBA. Interface inputs **shall** consist of the following:

- ICD Agreements/Understanding
 - Flight or test hardware supplied by one organization to the other organization (e.g., matching connectors, test fixtures, GSE)
 - Services and materials provided by one organization to the other organization (e.g., purge gas and fittings)
 - Flight hardware design and fabrication responsibility for interfaces (e.g., flight harness segments, ground straps, fasteners)
 - Pointing budgets and allocations, including estimated required spacecraft allocations
- ICD Design
 - First circuit schematics for all power and signal interfaces
 - Connector Input/Output (I/O) pinout tables (pin number and function) and harness wiring (e.g., twisted-shielded pair; 100 ohm controlled impedance)
 - CBE or measured orbit average power, by mode
 - CBE or measured Main Bus peak current
 - Modeled or measured Main Bus inrush transients
 - Modeled Main Bus impedance for source and load
 - Main Bus power service fault protection implementation (e.g., overcurrent set point)
 - Data – Rates (peak and orbit average), volume, compression, format, high speed data interface
 - Heater circuit schematic with equivalent resistance

-
- Design implementation of data bus and command/telemetry bus timing and protocols
 - Baseplate temperature range
 - Instrument calibration spacecraft requirements
 - The type, location, and orientation of the Instrument Alignment References
 - Mechanical (to be provided in separate Mechanical ICD appendix or document)
 - Mass, CG, MOI
 - All Instrument fields of view, including, where applicable, optical, glint, thermal cryo-radiator, electronics radiator, and solar diffuser fields of view.
 - Static and dynamic envelopes/volumes and keep out zones, including thermal blankets, Instrument connectors, mounting feet, Instrument-supplied harnesses, and unique mounts.
 - Information on field-of-view-related issues such as clearances, multi-path tolerances, indirect effects such as glint and allowable secondary radiation, and percent interference allowed.
 - EMI boresight stayout zones
 - Engineering drawings with a set of "not to exceed" dimensions for the launch, deployed on-orbit, and installation envelopes.
 - Specific requirements for pinning the Instrument.
 - The spacecraft mounting interface requirements for each separately-mountable Instrument component.
 - Mounting hole pattern
 - Mounting footprint
 - Mounting fasteners
 - Mounting type and materials (related to thermal and electrical bonding)
 - Volume/envelop with dimensions
 - Ground strap attach points, if any
 - Connector location with connector designators
 - Fields of View
 - Harness and GSE restraints, support, mounting interfaces
 - Purge gas interface
 - Location and proximity of other hardware mounted on the same panel
 - Instrument placement on satellite
 - Coordinate system
 - I&T access/clearances
 - Alignment features
 - Flight configuration (stowed for launch), to include with respect to fields of view and surrounding structure, components, and other hardware
 - On-orbit configurations (each state and intermediate positions of moving mechanical assemblies and deployables), to include with respect to fields of view and surrounding structure, components, and other hardware
 - Mounting hardware (and provider), torque specifications
 - Thermal
 - Interface temperatures, radiator FOV
 - Thermal Control System
 - FSW

- Protocol, rates, and handshaking (commanding acknowledgment)
 - Continuous, burst, etc.
 - FSW loads/patches for software updates
 - SMBA FSW updates
- Fault Monitoring
 - Spacecraft provided monitoring of Instrument faults
 - Responses to faults
- Operations
 - Operational modes: configuration, power draw, data rate, data volume, etc.
 - Calibration maneuvers
 - On-Orbit Constraints, Restrictions, and Warnings
- Ground support Equipment used during satellite I&T
 - Mechanical Fixtures
 - Lifting Slings
 - Simulators & Stimulators
- Contamination
 - Allocation of contaminant contributions from satellite sources
 - Satellite I&T and launch site facility constraints
 - MLI vent locations
 - Budget for particulate and molecular from delivery through EOL
 - Purge requirements (if any)
- Satellite level Instrument testing
 - Instrument simulators
 - I&T Constraints, Restrictions, and Warnings

The inputs to the ICD **shall** include:

- Special sequences required to recover from unannounced removal of power.
 - What activation sequences/events are required and where they are necessary.
- Specific spacecraft control requirements, including appropriate algorithms to be implemented.
- Include a list of items that may need to be accessible at satellite I&T, as well as a list of items to remove/install prior to flight, venting location, size, path, etc.
- The location, size, path, venting forces and impulses, direction of net thrust, and operation time of vents in the Instrument.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-12

2. Title:

FIELD PROGRAMMABLE GATE ARRAY (FPGA), APPLICATION SPECIFIC INTEGRATED CIRCUIT (ASIC), AND PROGRAMMABLE LOGIC DEVICE (PLD) CIRCUITWARE DESIGN DATA PACKAGE.

3. Reference:

GSFC 500-PG-8700-2.7B

4. Use:

An FPGA Design and Analysis Report is required for each FPGA design to allow Government review of Contractor FPGA, ASIC, and PLD designs. An ASIC Design and Analysis Report is required for each ASIC design to allow Government review of Contractor ASIC designs.

5. Preparation Information:

A Circuitware Design Data Package (CDDP) **shall** be delivered for each FPGA, ASIC or other PLD design (heritage, modified, or newly developed).

If an FPGA, ASIC or PLD is revised after its Design Data Package was delivered, then the Contractor **shall** deliver an updated Design Data Package so that each FPGAs, ASICs or PLDs final Design Data Package contains the flight design.

To facilitate useful and timely reviews, the Contractor may provide one or more interim CDDPs when a design is mature enough to benefit from independent review.

Note: For heritage or inherited FPGA, ASIC or PLD designs, it is expected that the items below will be delivered to the extent that these artifacts exist. It is not intended for the Contractor to create new artifacts for heritage or inherited designs simply for submission with this CDRL/DID.

The CDDP **shall** include:

- Design (place and route) database file(s) sufficient to open the implemented design in the FPGA, ASIC or PLD vendor's tool, typically as a .zip
- Post-route Verilog netlist (*.v) and timing (*.sdf)
- Synthesis report files
- Circuit board schematics containing this FPGA, ASIC, or PLD
- Circuit board netlist(s) (any ASCII format such as PADS, MGC, Allegro)

-
- Requirements, specifications, and verification document(s) as well as any supporting material (e.g., block diagrams, presentation material) relevant to the FPGA, ASIC, or PLD
 - Source code (e.g., VHDL, Verilog, state machines, state tables PDF of schematics) including any third party intellectual property code and cores
 - Responses to the GSFC 500-PG-8700-2.7B, Design of Space Flight Field Programmable Gate Arrays checklist or equivalent including:
 - Timing analyses for external inputs and outputs, internal domain(s), etc.
 - Disposition of all clock domain crossings.
 - Simulation analysis summary (e.g., functional coverage, code coverage) and simulation test bench/script code
 - Signal integrity analyses relevant to this FPGA, ASIC, or PLD
 - Power integrity analyses relevant to this FPGA, ASIC, or PLD

To facilitate reviewing the FPGA, ASIC, or PLD design, the CDDP may also include:

- System, box, and circuit board requirements, specifications, presentations, and verification document(s) relevant to the FPGA, ASIC, or PLD and its role in the system, box, and circuit board.
- Board part list (any ASCII or common spreadsheet format).
- PDF of the board layout, such as an assembly drawing.
- Any other files relevant to the review of the FPGA, ASIC, or PLD.

For any heritage, inherited, or build-to-print FPGA, ASIC, or PLD designs, the CDDP **shall** also include a “was/is” table consisting of:

- Full manufacturer part number
- Lot/Date code(s)
- Vendor part number (e.g., Altered Item Drawing number, etc.)
- Logic (e.g., any changes in the hardware description language code, etc.)
- Synthesis (e.g., not re-synthesized, re-synthesized to preclude Government Industry Data Exchange Program (GIDEP) xyz, etc.)
- Layout (e.g., not re-routed, re-routed per GIDEP xyz, etc.)
- Verifications
 - Simulations (e.g., not re-run, new simulations added, etc.)
 - Analyses (e.g., not updated, timing analysis updated with values from the latest software version that supports this FPGA, ASIC, or PLD, etc.)
 - Tests
- Environments
 - Operating temperature range
 - Operating voltage range(s)
 - Radiation
- Changes in manufacturing process and/or foundry

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-13

2. Title:

MASS PROPERTIES REPORT

3. Reference:

4. Use:

The purpose of the Mass Properties Report is to document all mass properties of the Instrument, its subsystems, and components from preliminary design through final assembly and satellite integration, and to satisfy the mass properties reporting requirements of the launch services provider.

5. Preparation Information:

This document is to provide a mass properties database for the Instrument and **shall** include:

- Mass, center of gravity, moments of inertia, products of inertia, uncompensated momentum, principal axis misalignment, relevant alignment measurements for each separately-mounted component of the Instrument, for each delivered unit, as well a composite total, and physical dimensions, where applicable.
- A report to be initially based upon calculated values.
- Updates as calculations are revised and actual measured data becomes available.

Following environmental testing and prior to the Instrument shipment to the satellite integrator, the report **shall** contain:

- A complete mass properties summary of the final Instrument mass properties as measured.
- The appropriate mass contingency for the current stage of hardware development, along with the allocated mass allowables.
- Documentation of movable masses, expendable masses, or deployables, if applicable, the respective variations **shall** be documented in the report.

The mass properties report **shall** contain the following:

- An overall Instrument mass summary.
- A detailed mass properties summary of all Instrument hardware organized by subsystem.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-14

2. Title:

SPARE PARTS PLAN AND LIST

3. Reference:

4. Use:

To review Contractor's spares plan and parts list.

5. Preparation Information:

The Spare Parts Plan **shall**:

- Define and justify the Contractor's position for the spares proposed for the Instrument.
- Present the schedule and method for obtaining the spares.
- Provide a listing of Spare Parts.
 - Document spares that are shared across all SMBA flight models and spares specific to a singular or subset of SMBA flight model(s).

For the purposes of this CDRL, the Contractor should concentrate on parts that are not commonly available or may cause schedule problems if out of stock.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-15

2. Title:ENGINEERING DEVELOPMENT PROGRAM
PLAN3. Reference:

NPR 7123.1, Appendix E

4. Use:

To detail the Engineering Development Program to achieve Technology Readiness Level (TRL) 6 (see NPR 7123.1, Appendix E) of all subsystems (example scan drive, digital processor, etc.), including hardware and software by CDR.

5. Preparation Information:

The Engineering Development Program Plan **shall**:

- Include a TRL assessment by assembly and subsystem (aggregation of component TRL levels is not sufficient to meet higher level assembly TRL requirements).
- Provide a general philosophy or approach to mitigating risk on the Instrument development through the use of engineering models.
- Provide justification for the proposed approach.
- Provide a list of all assemblies, subassemblies, or components of the Instrument which are planned to have engineering models.
- Discuss the functionality and purpose of each engineering model.
- Identify how the engineering model implementation fits into the Instrument development schedule.

2.4 Software Data Items

DATA ITEM DESCRIPTION

1. CDRL No.:

SW-1

2. Title:

INSTRUMENT SOFTWARE DEVELOPMENT
PLAN

3. Reference:

4. Use:

Defines Contractor activities required to develop and manage all software, whether developed by the Contractor or by a third party.

5. Preparation Information:

The Software Development Plan **shall** describe processes and activities used in the development and testing of the various types of software being acquired, acknowledging the fact that not all software must have the same criticality level or process requirements.

Topics in the Software Development Plan **shall** include:

- Project organizational structure showing authority and responsibility of each organizational unit, including external organizations.
- The safety criticality and classification of each of the systems and subsystems containing software.
- Engineering environment (for development, operation, or maintenance, as applicable), including test environment, library, equipment, facilities, standards, procedures, and tools.
- Life cycle processes and activities, including the software products, software services, non-deliverable items to be performed, budgets, staffing, acquisition approach, physical resources, software size, and schedules associated with the tasks.
- Management of the quality characteristics of the software products or services, to include malware prevention.
- Management of safety, security, privacy, and other critical requirements of the software products or services.
- Subcontractor management, including subcontractor selection and involvement between the subcontractor and the acquirer, if any.
- Verification and validation.
- Acquirer involvement.

- User involvement.
- Risk management.
- Security policy.
- Approval required by such means as regulations, required certifications, proprietary, usage, ownership,
- Process for scheduling, tracking, and reporting.
- Training of personnel, including project unique software training needs.
- Software life cycle model, including description of software integration and hardware/software integration processes, software delivery, and maintenance.
- Software and heritage software reuse plans
- Process for performing Unreachable Code Analysis and accepting and managing Unnecessary/Unreachable Code including risk to the mission associated with leaving the Unnecessary/Unreachable Code in the Flight Code.
- Configuration management.
- Software documentation tree.
- Software peer review/inspection process of software work products.
- Process for early identification of testing requirements that drive software design decisions (e.g., special system level timing requirements/checkpoint restart).
- Software metrics.
- Content of software documentation to be developed on the project.
- Management, development, and testing approach for handling any COTS, Government-off-the-shelf (GOTS), modified-off-the-shelf (MOTS), reused, or open source software component(s) that are included within a NASA system or subsystem.
- Software Assurance.
- Assurance Activities, Tools, and Techniques by discipline:
 - Software Quality (process and product)
 - Software Safety
 - Software Reliability
 - Software Verification and Validation
- Software Assurance Program Metrics.
- Problem Reporting and Corrective Action.
- Assurance records, collection, maintenance, and retention.

DATA ITEM DESCRIPTION

1. CDRL No.:

SW-2

2. Title:

SOFTWARE DETAILED DESIGN DOCUMENT

3. Reference:

4. Use:

For FSW and non-COTS GSE Software, the Software Detailed Design (SDD) Document describes the software design and operation for use by software maintenance team. It includes details of the architecture, structure, and organization of a particular CSCI, decomposing the top-level CSCI into CSC and lower levels of units as appropriate. The SDD describes each unit of software in terms of its interfaces (input/output), data architectures, and processing (e.g., logic, algorithms).

5. Preparation Information:

The SDD Document **shall** include details addressing the FSW as well and any custom (i.e. non-COTS) Software for GSE.

The SDD **shall** provide a system level design overview that contains:

- Design Methodology
- Design Overview
- Design Studies
- Design Issues
- Hardware Interface

The SDD **shall** provide a system design description that contains, at a minimum:

- Subsystem Description for each subsystem.
- Software Description for each software component:
 - Description of how the software item satisfies the software requirements, including algorithms, data structures, and functional decomposition.
 - Software item I/O description.
 - Static/architectural relationship of the software units.
 - Concept of execution, including data flow, control flow, and timing.
 - Requirements, design, and code traceability.
 - CSCI's planned utilization of computer hardware resources.
- Software Interface Control Description for both software-to-software and software-to-hardware interfaces.

The SDD **shall** describe the system operations design, including:

- Operations Scenarios.
- User-System Interface.
- Operations Environment and Facilities.

DATA ITEM DESCRIPTION

1. CDRL No.:

SW-3

2. Title:

VERSION DESCRIPTION DOCUMENT

3. Reference:

4. Use:

The Software Version Description Document (VDD) contains a letter describing what is being delivered, the actual executable software, and accompanying documentation. For all firmware media, formatted and numbered drawings are required with assembly numbers assigned to all loaded chips and drawing notes used to identify specific loading and test procedures, markings, etc.

5. Preparation Information:

The VDD **shall** include, as a minimum:

- Full identification of the system and software (e.g., numbers, titles, abbreviations, version numbers, and release numbers).
- Executable software (e.g., batch files, command files, data files, or other software needed to install the software on its target computer).
- Software life cycle data that defines the software product.
- Archive and release data.
- Instructions for building the executable software, including, for example, the instructions and data for compiling and linking and the procedures used for software recovery, software regeneration, testing, or modification.
- Data integrity checks for the executable object code and source code.
- Software product files (any files needed to install, build, operate, and maintain the software).
- Open change requests and/or problem reports, including any workarounds.
- Change requests and/or problem reports implemented in the current software version since the last Software Version Description was published.
- Confirmation that the software has been scanned for security defects and coding standard compliance.
- Software Bill of Materials.

DATA ITEM DESCRIPTION

1. CDRL No.:

SW-4

2. Title:INSTRUMENT SOFTWARE MAINTENANCE
PLAN3. Reference:

4. Use:

The Software Maintenance Plan **shall** contain the information required to sustain the software, including detailed procedures and functionalities. The Plan **shall** provide insight into the method, approach, responsibility, and processes to be followed for maintenance of software and its associated documentation.

5. Preparation Information:

The Software Maintenance Plan **shall** include, as a minimum:

- Plan information for the following activities:
 - Maintenance process implementation.
 - Problem and modification analysis.
 - Modification implementation.
 - Maintenance review/acceptance.
 - Migration.
 - Software Retirement.
 - Software Assurance.
 - Software Risk Assessment for all changes made during maintenance and operations.
- Specific standards, methods, tools, actions, procedures, and responsibilities associated with the maintenance process. In addition, the following elements are included:
 - Development and tracking of required upgrade intervals, including implementation plan.
 - Approach for the scheduling, implementation, and tracking of software upgrades.
 - Equipment and laboratories required for software verification and implementation.
 - Updates to documentation for modified software components.
 - Licensing agreements for software components.
 - Plan for and tracking of operational backup software (e.g., backup FSW, backup to the primary operational software).
 - Approach for the implementation of modifications to operational software (e.g., testing of software in development laboratory prior to operational use).

- Approach for software delivery process, including distribution to facilities and users of the software products and installation of the software in the target environment (including, but not limited to, spacecraft, simulators, Mission Control Center, and ground operations facilities).
- Approach for providing NASA access to the software version description data (e.g., revision number, licensing agreement).

DATA ITEM DESCRIPTION

1. CDRL No.:

SW-5

2. Title:

INSTRUMENT DATA SETS

3. Reference:

4. Use:

This data will be used to support ground system development and end-to-end mission rehearsals.

5. Preparation Information:

The sensor test data sets will be determined by the Contractor and **shall** include two sets: one representing normal Instrument operation and one with as many different configuration modes and data types as possible.

The Contractor **shall** prepare and deliver sensor test data sets simulating 72-hour continuous science data.

2.5 Integration and Test Data Items

DATA ITEM DESCRIPTION

1. CDRL No.:

IT-1

2. Title:

INSTRUMENT INTEGRATION, TEST, AND
VERIFICATION PLAN

3. Reference:

4. Use:

To provide information on the integration, testing, and verification of the Instrument, including the FSW. To provide an overview of the entire I&T flow and its relation to performance verification. To describe the methods, policies, and organization of the flight unit test program. To define the test requirements and the tests to be performed, instrumentation and locations, the tests methods, stimuli/measurements for acceptance criteria, and a time sequence for the testing. To define the FSW test (including acceptance test) plan including for new software capabilities for SMBA such as RFI detection and hyperspectral.

5. Preparation Information:

The Contractor **shall** provide an Instrument I&T Plan which includes:

- Descriptions of the series of activities required to integrate the various Instrument components into the final flight configuration.
- Descriptions of the methods, policies, and organization of the flight test program.
- Definitions of the test requirements and the tests to be performed, instrumentation and locations, the tests methods, stimuli/measurements for acceptance criteria, and a time sequence for the testing.
 - The Contractor may submit separate plans for each phase of the test program or one plan covering all phases.
- A list of all Contractor generated documents necessary to perform the integration and testing of the Instrument (e.g., the Environmental Verification Plan (EVP), the Environmental Test Matrix, requirements verification, etc.)
 - The EVP documents the Contractor's approach for environmental qualification and acceptance tests.
 - The Environmental Test Matrix summarizes the tests to be performed. The EVP and Environmental Test Matrix describes how the environmental verification complies with the requirements of the launch services provider.
- Cover the Engineering Development Program

- Cover all flight models while clearly identifying any differences in approach or activities between each of the subsequent flight models.
- Cover limitations in the Instrument verification program which preclude the verification by test of any system requirement. Examples include:
 - Inability to deploy hardware in a 1-g environment.
 - Facility limitations which do not allow testing at system level of assembly.
 - Inability to perform certain tests because of contamination control requirements.
 - Inability to perform powered-on testing because of voltage breakdown concerns.
- Evaluations of alternative tests and analyses and implemented, as appropriate, with an assessment of program risk to be included in the SPVP (CDRL SE-1).
- A flow of Instrument I&T sequence, including identification of calibration tests and characterizations.
- Cover contamination control at it applies to the integration and test program.

The I&T plan should contain the following information:

- Deliverables to Instrument I&T.
- Identify dependencies within the I&T sequence.
- Identify Instrument level requirements to be verified at each I&T stage, cross referenced to the System Performance Verification Plan (CDRL SE-1).
- Description of each Instrument activity or test including the general test configuration.
- Description of Instrument procedures required to support the activity or test.
- Description of simulated data/simulator requirements.
- Description of Instrument special handling requirements to support the activity or test.
- Description of Instrument ground support equipment requirements to support the activity or test.
- Description of Instrument safety requirements to support the activity or test.
- Description of Instrument personnel required to support the activity or test.
- Requirement to take photographs during the integration process for reference.
- Description of the process to be used for handling anomalies as they occur during I&T.
- Requirement of log books to be maintained throughout the I&T process.

This CDRL **shall** have a FSW Test Plan Appendix.

The FSW Test Plan Appendix **shall**:

- Define and describe the series of activities required to test, verify, and validate the FSW.
- Identify and describe all FSW tests that will be performed, including demonstrating hyperspectral capabilities and RFI detection capabilities.

DATA ITEM DESCRIPTION

1. CDRL No.:

IT-2

2. Title:

TEST RESULTS/REPORTS

3. Reference:

4. Use:

Report the results of all tests identified in the Test Plans, including test procedures used, test results, configuration status of all items under test, all measured data, a summary of observed results, and a discussion of compliance.

5. Preparation Information:

Test reports **shall** be delivered for all major environmental tests (TVAC, Dynamics, EMI/EMC) and performance verification tests (including the CPT). For all other tests, in lieu of reports, test results can be delivered in Contractor format. The results from all tests, including FSW tests, that are used for requirements verification **shall** be provided.

The following should be included in test reports at a minimum:

- Test identification and hardware configuration (and software version number if applicable) – for the specific test
- Facility description
- Reference – applicable test plan, test procedures, and test requirements, test log including the dates of the testing, photographs of test setup, any malfunction reports written during the test
- Test results, to include:
 - Identification of test results which confirmed the expected results as specified in the test plan/procedures or for which variations between actual and expected results were within specified tolerance. For the latter case, actual test results should be shown.
 - Identification of test results which differ from expected results beyond expected or acceptable limits.
 - Identification of any planned test objective or requirement for which actual results were not obtained. Reasons for not meeting the objective/requirement should be stated
 - Identification of any false or aberrant results noted during the test or subsequent analyses.
- Recommendations for subsequent actions should be stated, based on the test results.
 - Redesign of a particular component to enable the Instrument to meet a specific requirement which was not fulfilled.

- Revision of a system/subsystem specification in cases where the test results disclose ambiguity or conflicting requirements.
- The performance of additional tests where results were not acceptable to fulfill objectives.

DATA ITEM DESCRIPTION

1. CDRL No.:

IT-3

2. Title:DETAILED TEST PROCEDURES AND AS-RUN
TEST PROCEDURES

3. Reference:

4. Use:

To provide information on how the Instrument (system-level, hardware, software, etc.) and associated hardware such as cables and GSE/STE will be tested and how their performance will be verified.

For As-Run Procedures, to provide an archive of test procedures used during system development to aid in on-orbit anomaly resolution.

5. Preparation Information:

The Contractor **shall** provide detailed test procedures that:

- Are to be used during Instrument Level Testing (system-level, hardware, software, etc.) with the purpose and methodology, step by step instructions and processes, and identify the required environmental and configuration controls necessary for successful completion of the test clearly identified.
- Cover each test activity defined in the Performance Verification Plan (CDRL SE-1), including all Instrument test operations, interfaces, and Instrument performance requirements (i.e., electrical, structural and mechanical, EMC).
- Cover specialized tests such as mechanical function and deployments, environmental exposure tests (i.e., vacuum, vibration), Instrument calibration, GSE/STE calibration and checkout, and pre-launch end-to-end tests.
- Note: If tests are conducted in conjunction as part of a “group” test, for example, Limited Performance Test (LPT) or Comprehensive Performance Test (CPT), one test procedure may encompass this group.
- Indicate Data Quicklooks, Consent to Proceed (CTP), and Consent to Break (CTB) points associated with the test (as applicable).

As-Run Test Procedures **shall** be all as-run, red-lined, Contractor-format test procedures used during Instrument development, integration, and test at Instrument and at satellite levels. With the as-run procedures, include all deviations from the test procedures with a rationale for each deviation.

DATA ITEM DESCRIPTION

1. CDRL No.:

IT-4

2. Title:INSTRUMENT DELIVERY AND POST
DELIVERY SUPPORT (PDS) DOCUMENT

3. Reference:

4. Use:

This document is primarily intended to define the required Instrument operations from receipt at the spacecraft Contractor's facility through encapsulation. This document will include the details pertaining to equipment, personnel, testing, safety, and contamination providing the baseline integration plan, description, and procedure for conducting post-delivery activities from Instrument delivery at the spacecraft contractor to encapsulation for SMBA. This document will be used to guide the spacecraft and Instrument team to complete all operations and activities the Instrument requires to be ready for flight. This document will be used by the Government and spacecraft contractor to prepare a successful satellite integration and test campaign.

This document is also intended to cover packaging, handling, and transportation of the SMBA Instrument to the spacecraft vendor.

5. Preparation Information:

The Contractor **shall** provide the stated information below in sufficient detail to meet the intended uses of this document as stated above:

- Instrument GSE and equipment needed at the spacecraft and launch facilities: number and dimensions of racks, location of racks, phase of I&T when GSE is used, power requirements.
- All personnel required for all test events and all operations along with their roles and responsibilities.
- A mapping of all Instrument test activities mapped to each satellite level test event from Bench Acceptance Test (BAT) to launch site testing.
- Any safety concerns each operation presents, what personnel are required, constraints to be referenced, and if any special equipment is needed. Identify test flows and sequences that will involve hazards to personnel, flight hardware, and facilities that may impact any element of observatory integration and test.
- Cleaning and operational processes throughout all events at the spacecraft contractor's facility, in transport, and at the launch site. Additionally identify contamination sensitivities, sources, and concerns.

- Provide the Instrument scripts and inputs to spacecraft scripts necessary to run the Instrument after delivery.
 - Identify modifications to the flight telemetry limits (red/yellow limits) that are to be applied during the pre-launch phase.
- Identify red/green tag items (items to be removed or installed before flight) and when those items are to be removed/installed during the pre-launch phase.
- Plans, requirements, and procedures for maintaining and monitoring the Instruments while in storage at the spacecraft Contractor's facilities, and the actions to be taken in case of anomalies.
- Plans for periodic functional testing and maintenance while in storage (ex. exercising mechanisms).
- Plans, requirements, and procedures for packaging, handling, and transporting the Instruments to the spacecraft Contractor's facilities, and the actions to be taken in case of anomalies.

2.6 Mission Assurance Data Items

DATA ITEM DESCRIPTION

1. CDRL No.:

MA-2

2. Title:

ORBITAL DEBRIS ASSESSMENT REPORT
(ODAR) END OF MISSION PLAN (EOMP)

3. Reference:

NASA-STD-8719.14 Process for Limiting Orbital Debris

4. Use:

Provide information for development of the Orbital Debris Assessment Report (ODAR) and the End of Mission Plan (EOMP) deliveries per the content defined in NASA-STD 8719.14.

5. Preparation Information:

The Contractor **shall** provide the information necessary for the development of the ODAR and EOMP deliveries. NASA-STD-8719.14, Process for Limiting Orbital Debris, Appendix A (ODAR) and Appendix B (EOMP), provides details on what information is required for the project office to complete these analyses.

Note: Orbital Debris Assessment Software is available for download from Johnson Space Center at <https://orbitaldebris.jsc.nasa.gov/mitigation/debris-assessment-software.html>.

DATA ITEM DESCRIPTION

1. CDRL No.:

MA-3

2. Title:

SYSTEM SAFETY PLAN

3. Reference:

NPR 8715.7 Payload Safety Program
NASA-STD-8719.24 (with Annex), NASA Launch Vehicle Payload Safety Requirements

4. Use:

The Contractor **shall** prepare a System Safety Plan that describes the development and implementation of a system safety program that complies with the requirements of NPR 8715.7, the launch service provider, and launch range safety.

5. Preparation Information:The Contractor **shall**

- Define the roles and responsibilities of personnel.
- Define the required documentation, applicable requirements documents, and completion schedules for analyses, reviews, and safety packages.
- Address support for Safety Reviews, Safety Working Group Meetings, and Technical Interchange Meetings.
- Provide for early identification and control of hazards to personnel, facilities, support equipment, and the flight system during product development, including design, fabrication, test, transportation, and ground activities.
- Address compliance with the launch range safety requirements.
- Include a safety review process that meets the requirements of NPR 8715.7, Payloads Safety Program.
- Address compliance with safety requirements imposed by NASA and the Occupational Safety and Health Administration, address design and operational safety requirements, and address contractually imposed mission unique obligations.

DATA ITEM DESCRIPTION

1. CDRL No.:

MA-4

2. Title:SAFETY REQUIREMENTS COMPLIANCE
CHECKLIST

3. Reference:

NASA-STD-8719.24 (with Annex), NASA Launch Vehicle Payload Safety Requirements

4. Use:

The Safety Requirements Compliance Checklist indicates, for each requirement, whether the proposed design is compliant, non-compliant but meets intent, non-compliant, or not applicable. Any indication other than compliant requires rationale for acceptability.

The Annex is for project-level implementation, providing a template that can be tailored for each project's specific needs and is designed to be a starting point, allowing for a streamlined process of identifying and addressing project-specific risks and requirements.

5. Preparation Information:

The Contractor **shall** prepare a compliance checklist of all design, test, analysis, and data submittal requirements that includes the following:

- Criteria and requirement
- System
- Indication of compliance, noncompliance, or not applicable
- Rationale for indications other than compliant
- Resolution
- Reference
- Copies of Range Safety and NASA approved non-compliances, including waivers and equivalent levels of safety certifications.

Note: The Contractor **shall** submit safety waivers for non-compliant design elements per the SOW and CDRL SE-4.

DATA ITEM DESCRIPTION

1. CDRL No.:

MA-5

2. Title:INSTRUMENT SAFETY AND ASSESSMENT
REPORT (ISAR)3. Reference:

4. Use:

The Instrument Safety Assessment Report (ISAR) will identify safety features of the hardware, software, and system design as well as procedural, hardware, and software related hazards that may be present in the Instrument. This includes specific procedural controls and precautions that should be followed.

5. Preparation Information:

The ISAR **shall** include the following information:

- A comprehensive evaluation of the risks being assumed prior to the testing and operation of the Instrument.
- The safety criteria and methodology used to classify and rank hazards, including assumptions upon which the criteria or methodologies were based or derived.
- The results of hazard analyses and tests used to identify hazards in the system including:
 - Those hazards that still have a residual risk and the actions that have been taken to reduce the associated risk to a level contractually specified as acceptable.
 - Results of tests conducted to validate safety criteria, requirements, and analyses.
 - Hazard reports documenting the results of the hazard analyses to include a list of all significant hazards along with specific safety recommendations or precautions required to ensure safety of personnel, property, or the environment. Note: Identify whether or not the risks may be expected under normal or abnormal operating conditions.
 - The hazardous procedures and associated safeguards that will be used for integration and test activities and pre-launch activities.

The Preliminary hazard analysis, the hazard verification tracking log, and any hazardous procedures used during I&T or pre-launch processing at the launch site **shall** be completed in accordance with NASA-STD-8719.24 with Annex.

The Contractor **shall** submit with the final ISAR a signed statement that:

- Hazards have been closed out through elimination or through risk mitigations.
- Identifies hazards that have not been eliminated or controlled at the time of submission.

DATA ITEM DESCRIPTION

1. CDRL No.:

MA-6

2. Title:FAILURE MODES, EFFECTS, AND
CRITICALITY ANALYSIS (FMECA) AND
CRITICAL ITEMS LIST (CIL)3. Reference:

4. Use:

The FMECA Report **shall** explain the approach of the analysis, methodologies, assumptions, results, conclusions, and recommendations.

5. Preparation Information:

The FMECA Report **shall** include the following:

- Provide a ranking of items to determine criticality.
- CIL including item identification, cross-reference to FMECA line items. Appropriate retention rationale **shall** be stated. Rationale may include design features, historical performance, acceptance testing, manufacturing product assurance, corrective action recommendation/elimination of undesirable failure modes, proper design controls, and failure detection methods. The rationale also will contain data that describes operational constraints caused by occurrence of the failure and any measures that can be taken to restore the function on orbit where known.
- A discussion of the approach of the analysis, methodologies, assumptions, results, conclusions, and recommendations.
- Functional descriptions, functional block diagrams, and reliability block diagrams
- Equipment analyzed and data sources used
- Problems identified
- Provide an SPF list for failure modes. SPF analysis to include the root cause, mitigation, and retention rationale.
- Corrective actions
- Identification of failure modes, causes, criticality, and effects at the item, next higher level, and mission level, detection methods, and mitigating provisions.

DATA ITEM DESCRIPTION

1. CDRL No.:

MA-7

2. Title:

MASTER EEE PARTS LIST

3. Reference:

4. Use:

The Contractor **shall** maintain the Master EEE Parts List Information in a searchable electronic format. The GSFC Project Parts Engineer **shall** have access to the list.

5. Preparation Information:

The Contractor **shall** generate and maintain a Master Parts List with the minimum information listed below for the various stages throughout the project's life cycle:

- For EEE parts that are identified for potential flight use, the list **shall** include the following information:
 - Flight component identity to the circuit board level
 - Complete part number (i.e., Defense Supply Center Columbus part number, Specification Control Drawing part number, with all suffixes)
 - Manufacturer's Generic Part number
 - Manufacturer (not distributor)
 - Part Description (please include meaningful detail)
 - Federal Supply Class
 - Procurement Specification
 - Comments and clarifications, as appropriate
 - Estimated quantity required (for procurement forecasting)
- For EEE parts that are approved for flight use the list **shall** include the following information:
 - Procurement Part Number
 - Flight Part Number (if different from the procurement part number)
 - Package Style/Designation
 - PCB Status
 - PCB Approval Date
 - PCB Required Testing/Evaluations
 - Total Ionizing Dose (TID) rating

- Single Event Effects (SEE) rating
 - EEE derating information, and/or specifying derating to be evaluated in accordance with EEE-INST-002
- For EEE parts that are approved for as-designed use the list **shall** include the following information:
 - Assembly Name/Number
 - Next Level of Assembly
 - Need Quantity
 - Reference Designator(s)
 - Item number (if applicable)
- For EEE parts in flight hardware the list **shall** include the following information:
 - Assembly serial number
 - Item revision
 - Next Level of Assembly serial number
 - Lot/Date/Batch/Heat/Manufacturing Code, as applicable
 - Manufacturer's Cage Code (specific plant location when relevant)
 - Distributor/supplier, if applicable
 - Part number
 - Part serial number (if applicable)

DATA ITEM DESCRIPTION

1. CDRL No.:

MA-8

2. Title:MATERIALS IDENTIFICATION AND USAGE
LIST (MIUL)3. Reference:

4. Use:Provide a material list for materials used on the Instrument.

5. Preparation Information:

The Materials Identification and Usage List (MIUL) **shall** be prepared in accordance with NASA-STD-6016C section 4.1.2 with the following additions and prescribed changes to NASA-STD-6016C MIUL requirements:

The MIUL **shall**:

- Identify all unique materials and processes used in the construction of the deliverable hardware.
- Include thermal vacuum stability data, when appropriate, per NASA-STD-6016C, Section 4.2.3.6.
- Include the vacuum bake-out schedule for materials that fail outgassing requirements or provide a separate Materials Usage Agreement (MUA).
 - An MUA is for a material or process that does not meet the requirements of NASA-STD-6016 and does not affect reliability or safety when used.
 - The MUA includes the technical information required to justify the application.
 - MUAs for stress corrosion include a Stress Corrosion Cracking Evaluation Form per MSFC-STD-3029 (see NASA-STD-6016) and a stress analysis.
- Document, when appropriate, compliance to MSFC-STD-3029A.
- Clearly identify all materials that do not meet the requirements of NASA-STD-6016C.
- Include soldering flux and solvents used for cleaning flight electronic assemblies, other than isopropyl alcohol or deionized water.

The Contractor may use the GSFC outgassing database (URL: <http://outgassing.nasa.gov>) in addition to MAPTIS (URL: <http://maptis.nasa.gov>). The Contractor may use GSFC forms or the Contractor's equivalent forms in lieu of the MAPTIS format.

DATA ITEM DESCRIPTION

1. CDRL No.:

MA-9

2. Title:LIFE TEST PLAN AND FINAL REPORT FOR
LUBRICATED MECHANISMS

3. Reference:

NASA-STD-6016C Standard Materials and Processes Requirement for spacecraft
NASA-TM-86556 Lubrication Handbook for the Space Industry (Part A: Solid Lubricants, Part B: Liquid Lubricants)
NASA/CR-2005-213424 Lubrication for Space Applications

4. Use:

To describe the Life Test Plan and approach to qualifying all lubricated mechanisms either by life testing or heritage with an identical mechanism used in an identical application. To provide the approach and results of the lubricant loss analysis for all mechanisms to show that the design meets a 10X margin.

To document the life test final report at the end of each life test.

5. Preparation Information:

The plan **shall** contain:

- Description of lubricated mechanisms, performance functions, summary of subsystem specification, and life requirements.
- Heritage of identical mechanisms and descriptions of identical applications.
- Design, drawings, and lubrication system used by the mechanism.
- Test plan, including vacuum, temperature, and vibration test environmental conditions.
- Criteria for a successful test.

NASA-STD-6016C, Section 4.2.3.4, NASA-TM-86556, and NASA/CR-2005-213424 should be used as guides when developing the plan.

Alternatively, the Contractor **shall** provide detailed justification for NOT conducting life tests and/or lubricant loss analyses.

The life test final report **shall** be prepared in the Contractor's format.

DATA ITEM DESCRIPTION

1. CDRL No.:

MA-10

2. Title:END ITEM ACCEPTANCE DATA PACKAGE
(EIADP)3. Reference:

4. Use:

Provides a data package to include all paperwork necessary to assess the quality of the end item.

5. Preparation Information:

The Contractor **shall** prepare and deliver an End Item Acceptance Data Package (EIADP) for each Instrument and deliverable item in the contract including major subcontracted elements, and software elements. In general, major/critical suppliers are those who provide higher risk/complexity items, items of significant dollar and/or schedule value, or items near or on the critical path.

The Contractor prepares the EIADP as part of design development and implementation such that it is completed prior to delivery. The following items **shall** be included (as specified by requirement):

- The deliverable item name, serial number, part number, and classification status (e.g., flight, non-flight, ground support, etc.).
- Appropriate approval signatures (e.g., Contractor's quality representative, product design lead, Government Representative, etc.).
- List of shortages or open items at the time of acceptance with supporting rationale.
- Supplier list.
- As-built serialization.
- As-built configuration.
- In-process Work Orders (available for review at Contractor's – not a deliverable).
- Final assembly and test Work Order.
- Major MRB reports with a minimum:
 - Part number, serial number, problem description, disposition, root causes, and corrective actions.
- Major anomaly reports with a minimum:
 - Part number, serial number, problem description, disposition, root causes, and corrective actions.

- Acceptance testing procedures and report(s), including environmental testing.
- Quality Metric Trend data.
- As-built Master EEE parts list.
- As-built materials identification and usage list.
- Chronological history, including:
 - Total operating hours and failure-free hours of operation.
 - Total number of mechanical cycles and remaining cycle life.
 - Limited life items, including data regarding the life used and remaining.
 - As-built final assembly drawings.
 - Any PCB coupon results.
 - Photographic documentation of hardware (pre- and post-conformal coating for printed wiring assemblies, box or unit, subsystem, system, harness, structure, etc.).
- List or table of all waivers.
- Certificate of Compliance which is signed by management.

2.7 Calibration/Validation Data Items

DATA ITEM DESCRIPTION

1. CDRL No.:

CV-1

2. Title:

CALIBRATION/VALIDATION PLAN

3. Reference:

NIST Technical Note 1297 (1994 Edition) Guidelines for Evaluating and Expressing the Uncertainty of NIST Measurement Results
JCGM 100:2008(E) Evaluation of measurement data - Guide to the expression of uncertainty in measurement (2008) <https://doi.org/10.59161/JCGM100-2008E>

4. Use:

Document defines the planned calibration and validation (Cal/Val), intercalibration, RFI detection, and hyperspectral activities to be performed by the Contractor.

5. Preparation Information:

The Contractor **shall** work with the Government's Sensor Data Records Team to develop a Cal/Val Plan that describes the approach for calibrating, characterizing, and validating the spectral, spatial, radiometric, and geometric performance of the SMBA. The Cal/Val Plan **shall** address all requirements and tests related to science performance and Cal/Val, including intercalibration, RFI detection, and hyperspectral capabilities. This document **shall** include on-orbit Cal/Val requirements and references to the operations documents that describe these Cal/Val procedures. The Contractor's comprehensive radiometric/geometric calibration plan **shall** include preliminary work at subsystem levels of assembly as well as at the system level and component level (such as PRT sensor, Reflector emissivity).

The Cal/Val Plan **shall** incorporate the following information at a minimum:

- A description of planned tests and analyses including:
 - What is being tested or analyzed and how it relates to Instrument performance
 - The expected test or analysis results
 - Integration level for test or analysis model, i.e., part, subassembly, assembly, Instrument
 - Environmental conditions for test, e.g., ambient, thermal-vacuum, on-orbit
 - Sampling methods and their statistical validity
 - Operational phase of testing, i.e., pre-launch, or commissioning

-
- Theoretical basis for the test or analysis, i.e., how the test is performed, how the data are reduced and the method rationale supported by fundamental physics principles and equations)
 - The resolution, precision and accuracy of the results and relation to the expected results
 - Test configuration, i.e., equipment, test equipment calibration, and test setup
 - Description of test or analysis results usage, i.e., processing algorithms that use test or analysis results or calibration parameters generated by the test or analysis
 - A test and analysis schedule and flow chart
 - How test/analysis results are made available
 - Government access and participation in pre-launch testing including Government-owned standards or Government-performed Cal/Val-related characterization activities
 - Description of COTS and custom analysis tools
 - A description of the on-orbit calibration capabilities of the SMBA, including their design, characterization techniques, and operational implementation. This **shall** include a description of any necessary calibration data that must be collected during normal operations.
 - Reference Standards and their calibration traceability
 - Support data requirements
 - Tests to be conducted for calibrating the radiometric response of the Instrument
 - Inter-calibration (defined as calibration back to a NIST-traceable standard)
 - Tests to be conducted for intercalibrating the radiometric response of the Instrument to a NIST-traceable standard, including description of procedures and equipment
 - The method of intercalibrating any necessary secondary or transfer standards to a primary NIST-traceable standard.
 - An analysis of the uncertainties associated with the intercalibration process. Analyses **shall** follow NIST or International Bureau of Weights and Measures (BIPM) methodology such as described in NIST Technical Note 1297 (1994 Edition), Guidelines for Evaluating and Expressing the Uncertainty of NIST Measurement Results and JCGM 100:2008(E) Evaluation of measurement data - Guide to the expression of uncertainty in measurement (2008), online at <https://doi.org/10.59161/JCGM100-2008E>
 - Performed using Government Furnished Equipment (GFE) transfer standards or the Instrument provider's transfer standard
 - Validation of AP, RDR, TDR, and SDR products.

The Committee on Earth Observation satellites Working Group on Calibration & Validation provides the below definitions for Cal/Val. Reference <https://ceos.org/ourwork/workinggroups/wgcv/>.

- Calibration is the process of quantitatively defining a system's response to known and controlled signal inputs
- Validation, on the other hand, is the process of assessing, by independent means, the quality of the data products derived from those system outputs

Elaborating on these terms in the context of SMBA:

- Calibration inputs are electromagnetic energy from a scene at certain frequencies and the response (outputs) are the integrated energy under a channel spectral response function
- The data products have validation plans in CV-1 (i.e., assessed for quality) are:
 - APs
 - RDRs
 - TDRs
 - SDRs

The Instrument will produce APs, which are processed by the NOAA Ground System (NGS) to form RDR data. NGS will use the RDR, algorithms, and Instrument calibration data to produce higher-level data products such as TDRs and SDRs. A separate CV CDRL covers ground processing algorithm and software deliverables that will be key inputs for NGS processing from AP > RDR > TDR > SDR; therefore the CV-1 Cal/Val Plan needs to describe Contractor activities for validation of RDR, TDR, and SDR data. CV-1 does not need to describe Environmental Data Record (EDR) validation, as those processes are controlled by NOAA.

DATA ITEM DESCRIPTION

1. CDRL No.:

CV-2

2. Title:

CALIBRATION/VALIDATION PROCEDURES

3. Reference:

4. Use:

To verify that the planned Calibration, Validation, Intercalibration, RFI detection, and hyperspectral procedures meet the goals of the CV-1 Cal/Val and Intercalibration Plan.

5. Preparation Information:

For each test in the Cal/Val Plan, the Calibration, Validation or Intercalibration procedure for each test **shall** include:

- Test Objectives
- Test Methods
- Assumptions
- Applicable Documents and Software
- Associated algorithms
- Required Instrument Configuration, including any differences between pre-flight and flight configurations
- Mechanical and Electrical Test Equipment Configuration, including layout and interconnection of test equipment and articles including the grounding scheme. Location and identification of all measuring points on appropriate schematics and diagrams
- Test Equipment Identification
- Test Instrumentation
- Calibration reference requirements
- Safety Provisions and Cautions, including Identification of hazardous and potentially hazardous situations and operations and abort conditions
- Environmental and/or other conditions to be maintained, including contamination controls
- Personnel responsibilities and chain-of-command for test performance
- Expected results in telemetry and associated caution and warning levels.
- Program Quality Requirements
- Stimuli sources and associated and associated levels
- Test durations
- Test configurations
- Sequence of events

-
- Trending of performance characteristics during verification testing
 - Step-by-step instructions
 - Data Recording/Output Format Requirements
 - Data Recording Forms and Tables
 - Analysis techniques
 - Expected results
 - Pass/Fail Criteria
 - Test Termination Procedure

For each Analysis in the Cal/Val Plan, the procedure for each Analysis **shall** include:

- Analysis Objectives, i.e., short description of the item or items being validated by this analysis
- Analysis Methods including the analysis level of depth (i.e., 1st order, 2nd order, etc.)
- Assumptions
- Applicable Documents and Software
- Associated algorithms and software tools
- Required analysis mode/model configuration(s)
- List of input data that will be used and related test IDs that were used to acquire the data
- List of external reference sources of input data and/or model used in analysis
- List of fixed analysis parameters and set points
- List of adjustable analysis parameters and set points that may vary between runs.
- Equipment requirements
- Program Quality Requirements
- Sequence of events
- Step-by-step description of the data analysis procedure/algorithm
- Description of any statistical sampling method being used
- Data Recording/Output Format Requirements
- Data Recording Forms and Tables
- Expected results
- Pass/Fail Criteria

The procedures that govern antenna pattern measurement **shall**:

- Measure the far-field antenna radiation patterns
 - Note: Measurement of near-field antenna radiation patterns are preferred; however vendor and Government can coordinate if a model-only assessment of near-field is acceptable.
- Include beam pointing, -3dB beamwidth of cuts, beam efficiency (defined as the energy enclosed within 2.5 times the average 3dB beamwidth), and cross-polarization leakage.
- Be measured at the center frequency of the corresponding frequency band for analog frequency bands.
- Be measured at the two band edge frequencies and at the band center frequency for hyperspectral bands.
 - Include an additional two frequencies (one on each side midway between the band edge and the band center) for V-band.

- Include a minimum of four azimuth cuts to obtain two-dimensional radiation patterns:
 - Principal plane of the antenna (plane of symmetry of the antenna, also referred to as the down track plane).
 - A plane 90° from the principal plane (also referred to as the cross track plane).
 - 45° diagonal cut.
 - 135 diagonal cut.
- Measure each cut at two polarizations: co-polarized and cross-polarized.
 - Note: The co-polarization is parallel to the principal direction of polarization at the electrical boresight (beam center) of the beam position in question. Cross-polarization is orthogonal to the co-polarization.
- Measure the misalignment angle between feed and polarization separating component(s) for each frequency band.
- Characterize the cross-polarization antenna pattern with respect to scan angle.
- Characterize beam pointing error with respect to scan angle.
- Have a dynamic range of at least 80 dB.
- Have an amplitude accuracy less than +/- 3% in dB values throughout the dynamic range.
- Have a total elevation angular range of 360°, i.e., plus and minus 180° from electrical boresight.
- Have elevation angular intervals a maximum of one-fifteenth (1/15th) of the antenna Half Power Beamwidth.
- Have antenna pattern measurements at beam positions corresponding with:
 - Beginning, middle-left, nadir, middle-right, and end of the Earth-view sector.
 - Cold calibration reference beam position.
- Include accounting of gravity correction.
- Characterize the alignment of the antenna with respect to an Instrument reference.
- Evaluate antenna reflector emissivity through ground test or model simulation based on the surface roughness and dielectric properties of antenna reflector.

The procedures that govern Thermal Vacuum (TVAC) Calibration **shall**:

- Be written to perform end-to-end thermal vacuum calibration from the antenna through the Back End Electronics of all SMBA flight models.
- Produce the information required to evaluate the orbital measurement with a NEdT and nonlinearity better than the required values listed in the PRD.
- Emulate “test-as-you-fly” conditions.
- Collect calibration data in both Instrument operational mode and diagnostic mode.
- Emulate the mission thermal radiation environment.
- Perform a thermal transition test before the TVAC calibration cycle to investigate any stray light impacts on calibration targets.
- Provide a variable Earth scene target with a radiometric brightness temperature that varies from below 85K (goal 80K) to 330K in 25K increments; minimum 11 different scene temperatures.
- Set the Earth scene target within ± 1 K of its pre-selected value. The target temperature is determined from the weighted mean of temperature sensors embedded in it.

- Provide a cold space calibration target representative with a radiometric brightness no warmer than 85K (goal 80K). The target temperature is determined from the weighted mean of temperature sensors embedded in it.
- Maintain the indicated temperature change of each sensor taken individually to be less than 0.1 K during the period that calibration data is taken. If any sensor indicates a target temperature change in excess of 0.1 K, all data for that run will be discarded.
- Perform calibration at a minimum of three Instrument temperatures representing the SMBA Instrument low operational, mid operational, and high operational temperatures. Actual Instrument operational temperatures and control temperature locations will be agreed upon between the Contractor and the Government.
- Measure the Instrument internal temperature at one point within the Instrument where there is a flight temperature sensor. If the Instrument internal temperature varies by more than ± 0.5 K, all data for that run will be discarded.
- Perform hysteresis measurements at a minimum of two variable Earth scene target temperatures at each of the Instrument operational temperatures. Actual target temperatures will be agreed upon between the Contractor and the Government.
- Perform Point & Stare operations at a minimum of two variable Earth scene target temperatures at each of the tested Instrument temperatures to determine the Noise Power Spectral Density (NPSD) curves like that shown in Figure 1 below for each SMBA channel defined in PRD Table 4.1-2. The NPSD is an important characteristic of all radiometer channels relating quantities like NE Δ T, 1/f noise, gain variations, and striping. Actual target temperatures will be agreed upon between the Contractor and the Government.

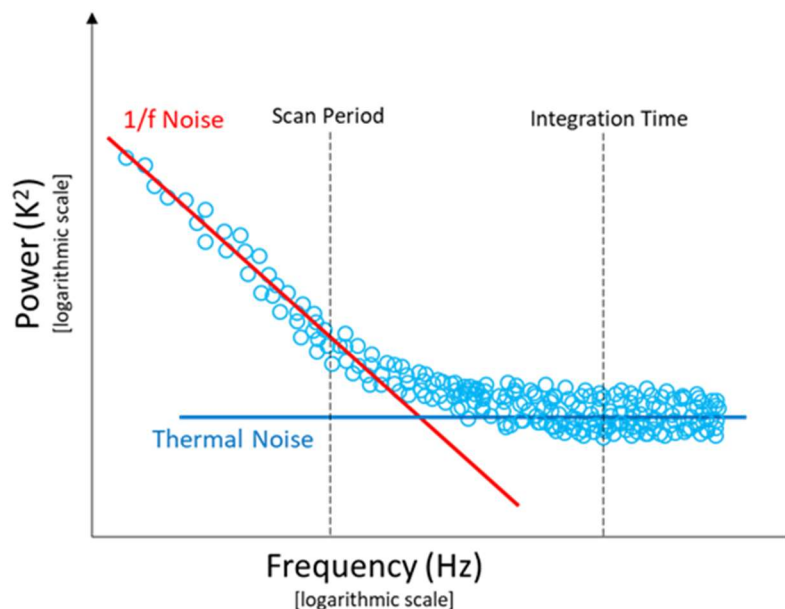


Figure 1. Noise Power Spectral Density Curves

DATA ITEM DESCRIPTION

1. CDRL No.:

CV-3

2. Title:CALIBRATION/VALIDATION RESULTS

3. Reference:

4. Use:

To provide results of individual calibration/validation tests and analyses. Of key importance is data to enable quantification of the main sources of uncertainty, and any applicable limitations on the test data, plus key assumptions made during the analysis.

5. Preparation Information:

The Contractor **shall** provide the Cal/Val and Intercalibration Results for each calibration, validation, or intercalibration test or analysis. All supporting data **shall** be provided in electronic format, where applicable.

Calibration, intercalibration, and data processing algorithms used **shall** be provided with the analysis of the test data. Test data **shall** include how the data was collected, the raw measurement data, analysis of data and any corrections to the data needed prior to use by data users, final data set after all corrections have been applied, and text to be associated with data for use.

The Cal/Val and Intercalibration Results **shall** include at a minimum:

- Identification of article or component tested or being validated with date of test or related analysis and the relevant Test/Analysis ID and run number.
- Performance trends during and between each planned test.
- List of uncertainty levels for each input data in analysis or test measurements.
- If a given test or analysis has more than one configuration, state the test or analysis configuration selected.
- Problems or failures with tests, procedures, or analyses.
- Anomalies and deviations from plans or procedures and their resolution/status.
- Test or analysis results, including:
 - Any calibration parameters to be used for on-orbit processing.
 - Comparison of results with expectations and requirements.
 - Interpretation of the results.
- General Reflector Antenna Software Package (or other agreed-upon software) models of near-field and far-field antenna radiation patterns, including comparison to as-measured patterns.

-
- Instrument Data Sets
 - General Notes
 - These data sets **shall** be provided on standard commercially available digital media.
 - These data **shall** be provided in a format accompanied by corresponding descriptive documentation.
 - For Instrument and satellite level testing where test pass/fail status reporting is required in a timely manner during the execution of the test (for example in TVAC before the consent to proceed to end a plateau), quick look data sufficient to evaluate the tests pass/fail status **shall** be provided in near real time.
 - For Instrument-level test output these data sets **shall** consist of:
 - Unprocessed detector sample data with associated time indexing information.
 - Instrument ancillary data that would be included in the SMBA mission data stream and associated time indexing information.
 - Instrument housekeeping information that would be included in the satellite telemetry stream and associated time indexing information.
 - Data sets at different levels of processing: APs, RDRs, TDRs, and SDRs. Reference CV-6 for Ground Processing Algorithm used to process the data.
 - For satellite-level test output these data sets **shall** consist of:
 - Unprocessed detector sample data with associated time indexing information.
 - Instrument ancillary data that would be included in the SMBA mission data stream and associated time indexing information.
 - Satellite ancillary data that would be included in the SMBA mission data stream and associated time indexing information. Ancillary data fields that are not available or are not meaningful in the prelaunch environment (e.g., ephemeris) may be simulated.
 - Instrument housekeeping information that would be included in the satellite telemetry stream and associated time indexing information.
 - Satellite housekeeping information that would be included in the satellite telemetry stream and associated time indexing information.
 - Data sets at different levels of processing: APs, RDRs, TDRs, and SDRs. Reference CV-6 for Ground Processing Algorithm used to process the data.

Characterization measurements of noise power spectral density **shall** be made for all continuity channels and provided in the Cal/Val reports.

DATA ITEM DESCRIPTION

1. CDRL No.:

CV-4

2. Title:RADIOMETRIC MATH MODEL
(RMM)/SENSOR PERFORMANCE MATH
MODEL (SPMM)3. Reference:

4. Use:

For modeling, evaluating, and characterizing the on-orbit end-to-end radiometric performance of the Instrument.

5. Preparation Information:

The terms “error” and “uncertainty” will be used interchangeably in this section.

The RMM/SPMM **shall** be used to: evaluate the end-to-end radiometric performance of the Instrument (Photons to calibrated brightness temperature); conduct sensitivity analyses; determine absolute and relative calibration accuracies; conduct error budget analysis for the calibration system, identify error contributors which can be eliminated or mitigated during the design phase; identify impact of error budget trades; assess Instrument performance in terms of Noise Figure (NF), NEdT, stability in orbit, etc., including radiometric sensitivity, calibration accuracy, nonlinearity, and noise; verify the emissivity of any on-board blackbody calibration references.

The model **shall** incorporate actual test and calibration data; the model **shall** be updated and refined during the course of the Instrument development program until it simulates Instrument performance within the accuracy required by the specifications.

The model **shall** include on-orbit calibration algorithms. The estimated data **shall** be replaced with measured data when it is available. The post-launch calibration algorithms are used along with pre-launch calibration data to demonstrate that the absolute and relative radiometric accuracies are being met. The calibration data **shall** also be provided in a mutually agreed upon electronic format. Algorithms to be provided **shall** include:

- Radiometric Temperature Computation
- Calibration Target Temperature Computation
- Physical Temperature Computation
- Brightness Temperature Computation

- Calibration Accuracy/Uncertainty Measurement
- NPSD and NEdT Measurements
- Non-Linearity
- Dynamic Range
- Gain Stability ($\Delta G/G$)
- Hot and Cold Calibration References
- Hot Target calibration reference uncertainty and corrections
- Cold Target calibration reference uncertainty and corrections
- Beam pointing error correction and Instrument mounting error correction
- Ground geolocation accuracy evaluation

The algorithm delivery **shall** consist of:

- An overview of each algorithm including the objective or purpose of the algorithm
- The rationale for each algorithm and identification of any specifications it supports
- A description of each algorithm's input parameters, mathematical equations, and resulting outputs, including the background and justification for their use, and a description of any external ancillary data (i.e., from the spacecraft) used by the algorithm
- A mathematical description of each algorithm
- An implementation of the algorithm in source code and identification of the Operating System that the source code was built on
- An error analysis showing the expected accuracy of the algorithm results
- A description of the methods used to validate each algorithm and description of the validation results
- A list of the Instrument data components used by each algorithm
- A list of the calibration parameters used by each algorithm
- A list of the output parameters from each algorithm
- A list of references

The SPM **shall** also include an error budget model consistent with the requirements for calibration accuracy in the PRD for assessment of propagate uncertainties in model input parameters to uncertainties in model output parameters, including error correlation effects. The basis for uncertainties in SPM inputs **shall** be documented. The delivered SPM **shall** include full documentation of the model, including the error propagation methodology, any software code/scripts to run the model, and instructions for Government customer use of the model.

System performance predictions **shall** include uncertainty estimates for calibrated radiance and equivalent blackbody temperatures based on propagation of uncertainties in SDR algorithm look-up table values and other SDR algorithm parameters.

DATA ITEM DESCRIPTION

1. CDRL No.:

CV-5

2. Title:

CALIBRATION DATA BOOK

3. Reference:

4. Use:

For demonstrating specification compliance and for full interpretation of the orbital data.

5. Preparation Information:

The calibration data book is to be prepared and processed for public release prior to delivery to NASA.

The calibration data book, which is to be delivered at the end of pre-launch Instrument calibration and intercalibration tests and as part of PSR Data Package, is to be a record of all pertinent test, calibration, and intercalibration data, including all raw data from which graphs, curves, etc., are prepared, and whose primary functions include: 1) providing fundamental calibration parameters for RDR, TDR, and SDR generation, and 2) providing users info needed to process data.

The calibration data book **shall** be updated after commissioning Cal/Val activities.

The calibration data book will provide fundamental calibration parameters needed by on-orbit calibration algorithm to produce TDRs and SDRs.

A summary section **shall** be prepared, which contains functional equations and charts pertinent to the data contained therein. The summary **shall** depict, as a minimum, all information pertinent to the performance requirements, the final system level bench test data, the final calibration and intercalibration of the Instrument and every Instrument subsystem as determined from all tests and calibrations performed prior to delivery of the Instrument to the Government, and a set of trend charts of all critical parameters.

Data **shall** be provided for each tested Instrument temperature.

The results of all tests and calibrations **shall** be summarized by the Contractor in the calibration log book(s).

The following information **shall** be derived from the calibration data:

-
- Temperature sensitivity (NEdT) of each channel
 - Standard deviation of individual calibration points from the best-fit calibration curves for each channel and each calibration
 - Calibration of any on-board calibration blackbodies in terms of effective brightness temperature versus temperature monitor raw outputs
 - Calibration curves of all temperature sensors.
 - Calibration curves of all voltage monitors.
 - Temperature stability data for the Instrument.
 - Beam profile in the far field for each channel
 - Non-linearity parameters
 - Antenna Pattern Data
 - Antenna sidelobe correction coefficients for cold reference target and antenna pattern correction coefficients for antenna temperature
 - Beam pointing error Euler angle information

Spectral Response Function (SRF) frequency data **shall** be with respect to the radio frequency (RF) at the antenna. This ensures that data is delivered with respect to the RF at the antenna, not with respect to any intermediate frequency (IF).

SRF data **shall** be delivered in digital format (such as .csv or MATLAB) that includes numerical values for gain vs. frequency and graphical plots of the SRFs. The data format is needed so that the Government can process and analyze SRF data.

SRFs, and associated text describing the SRFs, **shall** be prepared for public release and included in the calibration data book. SRFs are planned for public release to be made available to data users.

Pre-launch calibration parameters required for data processing **shall** be delivered in the calibration data book.

DATA ITEM DESCRIPTION

1. CDRL No.:

CV-6

2. Title:GROUND PROCESSING SOFTWARE

3. Reference:

4. Use:

To provide ground processing software that can be used to generate RDR, TDR, and SDR data products from SMBA and spacecraft APs. Accompanying the software will be a detailed description of the software and algorithms. This CDRL is an input to the NGS teams responsible for implementing the operational data processing code.

5. Preparation Information:The Ground Processing Software **shall:**

- Be prepared and processed for public release prior to delivery.
- Process Instrument and spacecraft APs into RDR, TDR, and SDR data products.
- Be complete and detailed, in Unified Modeling Language, such that a third party can generate executable code that correctly implements the algorithms.
- Include narrative descriptions of each routine, and include equations, figures, and other supporting information, where appropriate, in order to clearly convey the intent and context of the processing and enable software implementation.
- Include structural information such as hierarchy charts and data flow diagrams.
- Include algorithm descriptions and specifications including data interface definitions, flow charts, input and output variable lists, and pseudocode for each routine.
- Document any resampling algorithms.
- Demonstrate how to accommodate channel configuration re-programmability and RFI detection re-programmability
- Provide a complete description of all required datasets (e.g. initial gain and bias terms, detector maps) in Extensible Markup Language (XML).
- Describe the hardware used to test the algorithm and the measured performance of the algorithms on the test hardware (e.g., CPU and Memory utilization, Latency).

DATA ITEM DESCRIPTION

1. CDRL No.:

CV-7

2. Title:SPECTRAL RESPONSE FUNCTION (SRF)
MEASUREMENTS3. Reference:

PRD 4.9

4. Use:

To provide the SRF data of each channel.

5. Preparation Information:

This includes how the data was collected, the raw measurement data, analysis of data and any corrections to the data needed prior to use by data users, final data set after all corrections have been applied, and the text to be associated with data for use in CDRL CV-5, .

SRF data **shall**:

- Be with respect to the RF frequency at the antenna and include any corrections for pressure change from ground to altitude of satellite orbit.
- Be characterized at both ambient pressure and vacuum pressure for a minimum of three temperature conditions: coldest predicted operational temperature, warmest predicted operational temperature, and the median temperature between coldest and warmest predicted operational temperatures.
- Characterize SRF changes over the operational lifetime of the Instrument.
- Characterize the SRFs end-to-end during Instrument-level testing. If this is not feasible, the Contractor **shall** coordinate with the Government for testing at lower levels of assembly and/or analysis.
- Be updated when on-orbit channelization reprogramming results in new channels
- Have a minimum gain range of the achieved channel rejection magnitude per the PRD, and measure the noise floor. SRFs need to accurately represent the noise floor.
- Be delivered in digital format (such as .csv or MATLAB) that includes numerical values for gain vs. frequency and graphical plots of the SRFs.

SRFs, and associated text describing the SRFs, **shall** be prepared for public release and included in CDRL CV-5.

DATA ITEM DESCRIPTION

1. CDRL No.:

CV-8

2. Title:HYPERSPSCTRAL CAPABILITY TEST AND
ANALYSIS3. Reference:

4. Use:

To document the characteristics and performance of the SMBA Instrument's hyperspectral capability.

5. Preparation Information:

The delivery **shall** include:

- A description of the hyperspectral frequency configuration:
 - Identification of which bands and frequency ranges have hyperspectral capability.
 - The number of channels for each hyperspectral band.
 - The start and stop frequencies, the center frequency and bandwidth of each hyperspectral channel.
 - Performance trades and studies used to select the hyperspectral configuration, including rationale on the selected configuration.
- The performance characteristics of the hyperspectral channels:
 - The SRF of each channel, including a description of which passbands are combined to form the channel (a reference to CDRL CV-7 SRFs is acceptable).
 - The raw data used to derive the SRF along with the frequencies associated.
 - The description of the SRF measurement test that provides details about the test procedure (set up and instrumentation), the environmental conditions, the data generated, the data analysis and the results obtained.
 - A description of the algorithms that combine the channels including rationale; limitations and specifications; mathematical description; a performance analysis; and assumptions such as the data sets used for simulating the environmental conditions.
 - The start and stop frequencies, the center frequency stability, calibration uncertainty, and NEdT of each channel.
- Satellite and Mission-level considerations:
 - Flexibility in the frequency plan, including capabilities to update the frequencies pre- and post-launch
 - Power consumption impacts
 - Thermal impacts

- Data rate impacts on downlink
 - Data storage impacts
- Hyperspectral subject matter expert inputs:
 - SMBA block diagram of microwave radiometer signal chain and the description of the signal flow and digitization, including Fast Fourier Transforms (FFT), aggregating, windowing, and similar functions as applicable
 - A description of the key contributing hardware elements and their functions
 - A description of the software
 - Digitization errors, including how closely a digitized signal matches the analog signal
 - Frequency accuracy and stability
 - FFT output-to-frequency mapping
 - Choice of FFT window
 - Bandpass flatness
 - Aggregation algorithms to form channels

DATA ITEM DESCRIPTION

1. CDRL No.:

CV-9

2. Title:RADIO FREQUENCY INTERFERENCE
DETECTION CAPABILITY TEST AND
ANALYSIS3. Reference:

4. Use:

To document the capability for the SMBA Instrument to detect RFI from anthropogenic RF sources in the SMBA detector frequency range. Because the frequency range of the Instrument detector is near the range used for RF communications it is important for the Instrument to be able to identify RF interference that may contaminate science data measurements. This analysis documents the methods and capabilities the SMBA Instrument will use to identify RF interference.

5. Preparation Information:

The delivery **shall** consist of the description of the RF detection capability of the SMBA Instrument covering both the hardware and software aspects for each of the bands identified in PRD Table 4.1-2, Hyperspectral and RFI Detection Capabilities by Band. It **shall** include:

- The number of channels for each band
- The bandwidth of each channel
- The RFI detection algorithms chosen for each band
- The implementation of the RFI detection algorithms
- Any considerations overlapping hyperspectral and RFI configuration choices

For each of the selected RFI detection algorithm, the delivery **shall** consist of:

- An overview of each algorithm including the objective or purpose of the algorithm.
- The rationale for each algorithm and identification of any specifications and limitations.
- A description of each algorithm's inputs and outputs including any external ancillary data (i.e., thresholds) used by the algorithm.
- A mathematical description of each algorithm.
- An implementation of the algorithm in source code and identification of the Operating System that the source code was built on.
- A performance analysis of the algorithm that provides metrics on the algorithm including its detection sensitivity, its probability of detection, and its false alarm rate as function of RFI characteristics.

- A description of the methods used to validate each algorithm and description of the validation results.

At a minimum, RFI test cases **shall** include RFI algorithm capability assessment for each of the following categories of RFI:

- Continuous Wave with and without modulation.
- Pulsed signals with various duty cycles (5% ~ 60% duty cycle).
- Narrow band and wide band interference with various bandwidth (5-400MHz bandwidth interferes based on 5G Orthogonal Frequency Division Multiplexing spectrum and similar).
- Multiple sources with combinations of the three above.

All RFI test cases should be simulated and tested with a range of Interference to Noise Ratio (INR) in addition to a representative range of natural (RFI-free) background levels. For hard decision RFI detection algorithms (yes/no), Contractor **shall** provide the Receiver Operating Characteristic (ROC) curve where each point represents a different INR. For soft decision (selectable threshold), Contractor **shall** provide for each INR case a plot of the ROC curve where each point represents a different threshold.

A test report that describes the procedure used to test the SMBA Instrument's RFI detection capability for each of the bands identified in PRD Table 4.1-2, Hyperspectral and RFI Detection Capabilities by Band, **shall** be delivered. This test report **shall** include a description of:

- The RFI test set up and instrumentation
- The RFI characteristics tested
- The RFI detection algorithms tested
- The data sets generated by the test of the RFI detection algorithms
- Test Objectives
- Test Methods
- Assumptions
- Test Equipment Identification
- Environmental and/or other conditions to be maintained
- Test durations
- Test configurations
- Sequence of events
- Step-by-step instructions
- Data Recording/Output Format Requirements
- Data Recording Forms and Tables
- Analysis techniques
- Expected results
- Pass/Fail Criteria
- Test Termination Procedure

A test report that describes the procedure used for the RFI test cases reproducing what is currently observed from both terrestrial and space-based RFI sources **shall** be delivered. This test report **shall** include a description of:

- The RFI test set up and instrumentation
- The RFI characteristics tested
- The RFI detection algorithms tested
- The data sets generated for the test of the RFI detection algorithms
- Test Objectives
- Test Methods
- Assumptions
- Test Equipment Identification
- Environmental and/or other conditions to be maintained
- Test durations
- Test configurations
- Sequence of events
- Step-by-step instructions
- Data Recording/Output Format Requirements
- Data Recording Forms and Tables
- Analysis techniques
- Expected results
- Pass/Fail Criteria
- Test Termination Procedure

For each analysis required to validate the RFI detection capability tests, the procedure for each analysis **shall** include:

- Analysis Objectives, i.e., short description of the item or items being validated by this analysis
- Analysis Methods
- Assumptions
- Applicable Documents and Software
- Associated algorithms and software tools
- List of input data that will be used and related test IDs that were used to acquire the data
- List of external reference sources of input data and/or model used in analysis
- List of fixed analysis parameters and set points
- List of adjustable analysis parameters and set points that may vary between runs.
- Sequence of events
- Step-by-step description of the data analysis procedure/algorithm
- Data Recording/Output Format Requirements
- Data Recording Forms and Tables
- Expected results
- Pass/Fail Criteria

Raw data generated to assess the SMBA Instrument's RFI detection capability for each test performed **shall** be delivered along with the test report and the analysis report. The raw data delivery **shall** include:

- The ancillary data used for the test
- Raw data

- Frequency channels, their bandwidth
- Implementation of the RFI detection algorithms tested

2.8 On-Orbit Data Items

DATA ITEM DESCRIPTION

1. CDRL No.:

OO-1

2. Title:

INSTRUMENT ON-ORBIT COMMISSIONING
PLAN

3. Reference:

4. Use:

To define how the Instrument will be activated, calibrated, and tested (functionally and performance), and how its on-orbit performance will be verified. The Plan should include test activities, test equipment and plans for inclusion of the Government Cal/Val team, and any other relevant Government partners.

5. Preparation Information:

This CDRL item may be combined with OO-5.

The On Orbit Commissioning Plan **shall** be the governing document for initialization and validation of the Instrument during the pre-operational check-out period. The plan **shall** include:

- A summary of the initialization and verification methodology.
- A matrix or list of any Instrument requirements, with associated acceptance criteria, to be verified on-orbit which is cross-referenced to the appropriate Instrument On-Orbit Test Procedures or Cal/Val Procedures.
- Notional schedule of initialization and verification activities including duration estimates.
- Constraints to operations.
- A list of required equipment.

DATA ITEM DESCRIPTION

1. CDRL No.:

OO-2

2. Title:INSTRUMENT USER'S MANUAL

3. Reference:

4. Use:

The manual **shall** be a self-contained document in that a reader not familiar with the Instrument can obtain a reasonably complete understanding of the Instrument without recourse to another document or drawing. The document is not meant to be an engineering working document but a reference document for SMBA Instrument data users including: satellite operators, Government personnel, scientists, spacecraft Contractor personnel, and the general public.

5. Preparation Information:

The Instrument User's Manual **shall** provide a non-proprietary description of the system, subsystems, functions, and operations (with emphasis on how the system operates), with illustrations, block diagrams and circuitry descriptions. The Instrument to spacecraft interface **shall** be described. Physical details (mechanical, electrical, thermal, optical train & detectors, etc.), information on the software design, interfaces, Instrument modes and states, commands and telemetry, and performance characteristics **shall** be provided by inclusion or by reference. The report **shall** be a self-contained document in that a reader not familiar with the Instrument can obtain a reasonably complete understanding of the Instrument and its operation without recourse to another document or drawing. The document is not meant to be an engineering working document but a reference document for Government personnel, scientists, spacecraft Contractor personnel, and the general public of SMBA Instrument data users (including international users).

The SMBA Instrument User's Manual **shall** characterize Instrument performance with respect to: relative spectral response; radiometric accuracy, sensitivity, and stability; and line-of-sight accuracy. Each Manual **shall** provide pre-flight test results characterizing performance and a flight performance evaluation with updates at major milestones. The document **shall** describe the Instrument modes of operation and the equipment, methods, accuracies, and concepts of operation for in-flight calibration of radiometric response and line-of-sight. The document **shall** include a description of the data format and packet structure.

In general, if there is any doubt whether an operator may need to know the information, the Contractor should include it.

Constraints **shall** be discussed as appropriate and in context; e.g., if critical commands must be enabled prior to sending, this should be discussed in the command and data handling section.

The User's Manual **shall** cover the life of the Instrument from launch pad through decommissioning.

Note the following is not intended to be a table of contents. The information below does not have to be in the order or groupings given here, but are provided to ensure expected content is included in the deliverable.

Information **shall** include, but is not limited to, the items below:

- Sensor Overview, including science mission:
 - Channel Wavelengths/Frequencies and Resolutions
 - Field of View (FOV)/footprint on the Earth
 - Scan period
 - Scan patterns
 - A description of the hyperspectral capabilities
 - For hyperspectral, include:
 - Identification of which bands and frequency ranges have hyperspectral capability
 - The number of channels for each hyperspectral band
 - The start and stop frequencies, the center frequency and bandwidth of each hyperspectral channel
 - Performance trades and studies used to select the hyperspectral configuration, including rationale on the selected configuration
 - The SRF of each channel, including a description of which passbands are combined to form the channel (a reference to CDRL CV-6 SRFs is acceptable)
 - The raw data used to derive the SRF along with the frequencies associated
 - The description of the SRF measurement test that provides details about the test procedure (set up and instrumentation), the environmental conditions, the data generated, the data analysis and the results obtained
 - A description of the algorithms that combine the channels including rationale; limitations and specifications; mathematical description; a performance analysis; and assumptions such as the data sets used for simulating the environmental conditions
 - The center frequency stability, calibration accuracy, and NEDT of each channel
 - Flexibility in the frequency plan, including capabilities to update the frequencies pre- and post-launch
 - Power consumption impacts
 - Thermal impacts
 - Data rate impacts on downlink
 - Data storage impacts
 - Description of the signal flow and digitization, including FFTs, aggregating, windowing, and similar functions as applicable

- A description of the key contributing hardware elements and their functions
 - A description of the software
 - Digitization errors, including how closely a digitized signal matches the analog signal
- A description of the RFI capabilities
- For RFI include:
 - The Instrument bands that have RFI capability
 - The number of channels for each band
 - The bandwidth of each channel
 - The RFI detection algorithms chosen for each band
 - The implementation of the RFI detection algorithms
 - An overview of each algorithm including the objective or purpose of the algorithm
 - The rationale for each algorithm and identification of any specifications and limitations
 - A description of each algorithm's inputs and outputs including any external ancillary data (i.e., thresholds) used by the algorithm
 - A mathematical description of each algorithm
 - An implementation of the algorithm in source code and identification of the Operating System that the source code was built on.
 - A performance analysis of the algorithm that provides metrics on the algorithm including its detection sensitivity, its probability of detection, and its false alarm rate as function of RFI characteristics
 - A description of the methods used to validate each algorithm and description of the validation results
 - Any considerations overlapping hyperspectral and RFI configuration choices
- Subsystem/Component Overview (the purpose of this section is to inform the reader of overall subsystem information; however it should cover in detail those aspects that are monitored or changeable by the operator):
 - Mechanical
 - Include diagrams of motors and information about bearings and lubricants, because electro-mechanical devices are notorious troublemakers
 - Electrical
 - Nominal operating power ranges
 - Thermal
 - Locations of all heaters and their relationship (location relative) to thermistors
 - Heater set points
 - Nominal operating temperature ranges
 - Optical/RF/energy path through the system
 - Detector layout (if appropriate)
 - Calibration targets/sources
 - FSW

- Software summary, including: application, inventory, environment, organization, overview of operation, contingencies, alternate states, and modes of operation.
 - Functional description
 - Simplified flowchart
 - Fault Protection
 - What it monitors
 - What it does in response to ‘violations’ of monitored information
 - Description and discussion of uploadable and downloadable tables including how to know when an update is needed
 - Fault codes (or where to find)
 - Response to fault codes (or where to find)
- Sensor Modes
 - Description of and how to recognize each mode
 - Allowable Transitions
 - Recommended Transitions
 - Indicate what transitions can occur autonomously versus by command
 - Table of major components and their status (e.g., ON/OFF) for each mode
 - Table of telemetry available in each mode
 - Activities allowable in each mode (e.g., loads and/or dumps)
- Flight model-specific signatures
 - ‘Features’ of the specific flight model being documented
 - Differences with existing/operating sensors should be explicitly pointed out
- Interfaces
 - Sensor Internal
 - Spacecraft External
 - Point-to-point telemetry and commands
 - Survival heater locations, set points and relationship to thermistors (if not covered under sensor thermal subsystem)
 - Databus interaction
 - Timing, clocks, and synchronization
 - Average and peak data rate
 - Average and peak power consumption
- Command and Data Handling
 - APIDs
 - Telemetry Rates
 - Packet descriptions
 - Description on on-board stored commanding and macro capabilities
 - Critical commands
 - Database
 - Flight Telemetry Limits ‘philosophy’: i.e., How did the Contractor ‘choose’ the flight limits? E.g., Did the Contractor take the ‘tested to’ maximum and add margin? Did the Contractor take the ‘Qual’ limits and add margin? If a limits philosophy requirement is defined elsewhere, it supersedes this guidance.

- Database information **shall** include Mission Data Format file schema that includes all relevant field metadata, deliverable in excel or similar format.
- Science
 - Algorithms
 - On-board data processing
- Overview of Recommended Operating Procedures (ROP)
 - Purpose and short summary of functionality
- Normal Operations
 - On-Orbit Activation Sequence (could be a ROP or series of ROPs) including timing as determined from testing, e.g., cool-down is expected to take x minutes or x hours or x days
 - Cal/Val Activities
 - On-Orbit Maintenance Activities
 - Constraints/configurations for Maneuvers DELTA-V and DELTA-I maneuvers
 - Trending Recommendations (health and safety and science performance parameters)
 - Nominal values
 - Frequency of trending
 - Limits
 - Special Activities (e.g., Cal/Val maneuvers, mid-mission out-gassing)
 - Table and/or FSW Loads
 - Hyperspectral and RFI related configuration changes
- Contingency Operations and Troubleshooting
 - Red/yellow limit responses
 - Document common or ‘expected’ anomalies and their response
 - Redundancy and redundancy management
 - Diagnostic modes/activities/data and when appropriate to use

DATA ITEM DESCRIPTION

1. CDRL No.:

OO-3

2. Title:

COMMISSIONING REPORT

3. Reference:

4. Use:

The purpose of the Commissioning Report is to show that the on-orbit performance meets the specifications.

5. Preparation Information:

For each activity required to validate on-orbit performance, the following information **shall** be reported:

- Activity information including name, short description, and analysis plan.
- Success criteria.
- Test data used for validation including test date(s) and time.
- Activity results, including activity date and time, and data used for validation on the same level as the TVAC reports; where possible show how on-orbit performance compares to ground characterization data.
- Any test anomalies or idiosyncrasies and their impacts.
- Health, functionality, and performance.

DATA ITEM DESCRIPTION

1. CDRL No.:

OO-4

2. Title:COMMAND AND TELEMETRY HANDBOOK
AND COMMAND AND TELEMETRY
DATABASE (CTDB)

3. Reference:

4. Use:Establishes the command and telemetry definitions for the SMBA.

5. Preparation Information:The Command and Telemetry Handbook **shall** be delivered in Contractor format.The Handbook **shall** contain the telemetry information. This section of the Handbook **shall** provide at least, but not be limited to, the following:

- Detailed listing of all telemetry assignments and Parameter IDs.
- Key parameters necessary for description of the telemetry requirements as a part of the list.
- Summary quantifying the number and type of telemetry assignments for each subsystem and the number of spares remaining.
- Description of telemetry interfaces, telemetry format, telemetry requirements data, and caution and warning (red and yellow) limits.
- Listing of telemetry assignments that confirm commands.
- Data format, to include: detailed bit definitions, command, meta and ancillary data definitions for all telemetry and command data (command and telemetry list).

The Handbook **shall** contain the command information. This section of the Handbook **shall** provide at least, but not be limited to, the following:

- Detailed listing of all serial and digital commands.
- Key parameters necessary for description of the commands as part of this listing.
- Summary quantifying the number of types of commands used by each subsystem and by each unit and the number of spares remaining.
- Description of command input, command verification, command rate and filler (no-op) commands.
- Description of command requirements data and information necessary for interpretation.
- Listing of commands verified by telemetry and the telemetry verifiers.

- Schematic reference for each command.
- Identification of hazardous commands (that can cause damage to the Instrument, spacecraft, or personnel).

The Handbook **shall** contain information and definitions for all on-board configurable FSW parameters, including on-board FSW table information and table definitions.

The Instrument Command and Telemetry Database (CTDB) **shall** be provided in a format agreed to with the spacecraft vendor and operations center.

The CTDB **shall** contain:

- All commands required to operate, monitor, and control the SMBA.
- All telemetry parameters required to operate, monitor, and control the SMBA.
- Instrument command and telemetry unique identifiers.

DATA ITEM DESCRIPTION

1. CDRL No.:

OO-5

2. Title:INSTRUMENT LAUNCH AND ON-ORBIT
OPERATIONS AND CONTINGENCY
PROCEDURES

3. Reference:

4. Use:

The Launch and Early Orbit Procedure section is required to coordinate the launch operation.

The Instrument On-Orbit Procedures section contains the complete set of procedures required for operating the Instrument following completion of commissioning activities. It **shall** cover all mission phases: activation & checkout, calibration/validation, routine operations, and contingencies.

The Instrument On-Orbit Test and Cal/Val Procedures section of the document contains the set of procedures required to conduct activation, testing, and calibration/validation during the commissioning phase of the mission. These procedures are unique to the activation and commissioning phase and do not duplicate the procedures in the Instrument On-Orbit Operations and Contingency Procedures section. This document **shall** include the plans, requirements, and procedures for on-orbit sensor calibrations.

5. Preparation Information:

The Instrument Launch and Early Orbit Procedure section of the document **shall** contain Instrument sequences required to support the Instrument state of health monitoring and operational events during the launch phase of the mission. These procedures are applicable from launch vehicle-satellite separation until the SMBA satellite is configured for on-orbit operation prior to the start of the on-orbit system test activities.

The procedure **shall** include:

- Detailed flight timeline.
- Survival heater configurations (when they need to be turned on), set points, etc.
- Script of each communication event including required actions and responses.
- Launch commitment criteria.

The Instrument On-Orbit Procedures section of the document **shall** provide a detailed set of operations procedures for operating the Instrument. It **shall** provide detailed operating

procedures for each Instrument mode, including overview/purpose, prerequisites, applicable mission phases, affected systems/subsystems, flowcharts, commands/mnemonics/parameter values, telemetry mnemonics/verification values, constraints (e.g., time-spacing between commands) and cautions, associated contingencies, duration (if known) and start time (if known). These procedures **shall** include:

- Normal on-orbit command and control operations
- Instrument State-of-Health Monitoring and management
- Instrument mode transition and mode operations
- Contingency and recovery procedures
- Calibration procedures
- Instrument procedures for satellite reconfigurations

The Instrument On-Orbit Test and Cal/Val Procedures section of the document **shall** provide a detailed set of operations procedures for activating, testing, and Cal/Val of the Instrument.

These procedures **shall** include:

- Instrument command and control operations
- Instrument State-of-Health Monitoring and management
- Instrument mode transition and mode operations
- Contingency and recovery procedures
- Calibration procedures
- Instrument procedures for satellite reconfigurations

Detailed Cal/Val Procedures delivered under CDRL CV-2 may be referenced from this document instead of duplicated.

All sections, all procedures **shall** include:

- Test/Procedure Purpose
- Any Instrument requirement(s) being verified
- Procedure Methodology
- Support Resources Required
- Satellite and Instrument configuration before and after the procedure is executed
- Step-by-step commands to be issued and expected Instrument response after each step, including possible contingencies and responses
- Cautions and warnings
- Procedure prerequisite events and conditions
- Relevant operational constraints

DATA ITEM DESCRIPTION

1. CDRL No.:

OO-6

2. Title:INSTRUMENT SIMULATOR USER'S GUIDE

3. Reference:

4. Use:

The Simulator User's Guide describes the operation of simulator for use by the spacecraft Contractor

5. Preparation Information:

The Instrument Simulator User's Guide **shall** contain the information required to use the simulator, including detailed procedures and functionalities. The User's Guide **shall** show all simulator Graphical User Interfaces (GUI) and detail the usage of each GUI. It **shall** give detailed descriptions of major functionality provided by the simulator, and give step-by-step instructions on how to install and use the simulator to achieve these functionalities. The Simulator User's Guide **shall** detail the various modes of operation based on access control and show the difference in screen activations based on a user's input.

The Simulator User's Guide **shall**:

- List all alerts or notifications produced by the simulator along with their meanings.
- Indicate how to start the simulator.
- Indicate recovery methods in cases of irrevocable errors or faults.
- Indicate data types and command arguments expected within each field of each GUI.
- Indicate limitations and differences from the flight Instrument.
- If necessary, describe any unique configurations or commands associated with each instantiation of the simulator.
- Include minimum hardware platform and operating system requirements.
- Include facility requirements (number of racks, mass, power, footprint, etc.).
- Include simulator external interfaces to the NEON Series-1 spacecraft simulator – electrical, command/telemetry, data rates, protocol, packet structure/definition.
- Include modes of operation of the Simulator.
- Include simulator displays.
- Include access control requirements.
- Include simulator input format, file, and data structure.

Appendix A: Abbreviations and Acronyms

Abbreviation/Acronym	Definition
$\Delta G/G$	Gain Stability
A	Approval
ACS	Attitude Control System
ADB	Ada body file extension
AFM	Additional Flight Models (i.e., in addition to FM1)
AM	Additive Manufacturing
AMCP	Additive Manufacturing Control Plan
ANSI	American National Standards Institute
AP	Application Packet
APID	Application Process Identifier
APPLIC	Applicability
ASCII	American Code Standard for Information Interchange
ASIC	Application Specific Integrated Circuit
ATMS	Advanced Technology Microwave Sounder
ATP	Ambient Temperature and Pressure
ATP	Authority To Proceed
AVI	Audio Video Interleave
BAT	Bench Acceptance Test
BCR	Baseline Change Request
BCWP	Budgeted Cost of Work Performed
BOL	Beginning of Life
CA	Control Account
CADRe	Cost Analysis Data Requirement
CAD	Computer-Aided Design
CAE	Computer-Aided Engineering
Cal/Val	Calibration/Validation
CAT	Category
CBE	Current Best Estimate
CCB	Configuration Control Board
CCP	Contamination Control Plan
CCR	Configuration Change Request
CDDP	Circuitware Design Data Package
CDR	Critical Design Review
CDRL	Contract Data Requirements List
CG	Center of Gravity
CIIL	Configuration Item Identification List
CIL	Critical Items List
CLIN	Contract Line Item Number
CM	Configuration Management
CMO	Configuration Management Office
CMP	Configuration Management Plan
COR	Contracting Officer's Representative
COTS	Commercial-off-the-shelf
CPD	Contract Performance Dataset

CPR	Contract Performance Report
CPT	Comprehensive Performance Test
CPU	Central Processing Unit
CRC	Cyclic Redundancy Check
CrIS	Cross-track Infrared Sounder
CSC	Computer Software Components
CSCI	Computer Software Configuration Item
CSU	Computer Software Unit
csv	Comma Separated Values file extension
CTB	Consent to Break
CTDB	Command and Telemetry Database
CTI	Consent to Integrate
CTP	Consent to Proceed
CV	Calibration/Validation
CWBS	Contract Work Breakdown Structure
D	Definitization
DACA	Days After Contract Award
DC	Direct Current
dCDR	Delta Critical Design Review
DD	Department of Defense (forms only)
Del	Deliverable
DEV	Development
DFMR	Designed For Minimum Risk
DID	Data Item Description
DM	Data manager
DMR	Design Modification Review
DPA	Destructive Physical Analysis
DOORS	Dynamic Object-Oriented Requirements System
DRD	Data Requirements Document
DTD	Document Type Definition
DTM	Detailed Thermal Model
DWF	Design Web Format
E	Electronic Format
EAC	Estimate at Completion
EAR	Export Administration Regulations
ECR	Engineering Change Request
EDR	Environmental Data Records
EDU	Engineering Development Unit
EEE	Electrical, Electronic, and Electromechanical
EEO	Early Engineering Opportunity
EGSE	Electrical Ground Support Equipment
EIADP	End Item Acceptance Data Package
EIDP	End Item Data Package
EMC	Electromagnetic Compatibility
EMI	Electromagnetic Interference
EOL	End of Life
EOMP	End of Mission Plan

EPR	Engineering Peer Review
ERB	Engineering Review Board
ESD	Electrostatic Discharge
ESRD	Electrical System Requirements Document
ETU	Engineering Test Unit
EVMIG	Earned Value Management Implementation Guide
EVM(S)	Earned Value Management (System)
EVP	Environmental Verification Plan
EVT	Earned Value Technique
FAR	Federal Acquisition Regulation
FCA	Functional Configuration Audit
FFS	File Format Specification
FFT	Fast Fourier Transform
FM	Flight Model
FMEA	Failure Modes and Effects Analysis
FMECA	Failure Modes and Effects Criticality Analysis
FOV	Field of View
FPGA	Field Programmable Gate Array
FQT	Formal Qualification Test
FRB	Failure Review Board
FRR	Flight Readiness Review
FSTB	Flight Software Test Bed
FSW	Flight Software
FVTS	Flight Vehicle Test Suite
G-R	Giver-Receiver
GFE	Government Furnished Equipment
GIDEP	Government Industry Data Exchange Program
GOLD	GSFC Open Learning Design
GOTS	Government-off-the-shelf
GPR	Goddard Procedural Requirement
GSE	Ground Support Equipment
GSFC	Goddard Space Flight Center
GUI	Graphical User Interface
H	Hardcopy
H&S	Hardness and Survivability
Hz	Hertz
I	Information
I/O	Input/Output
I&T	Integration and Test
ICDR	Instrument CDR
ID	Identification
IBR	Integrated Baseline Review
ICD	Interface Control Document
ICDR	Instrument Critical Design Review
iDel	Internal Deliverable
IEEE	Institute of Electrical and Electronics Engineers
IF	Intermediate Frequency

IMS	Integrated Master Schedule
INR	Interference to Noise Ratio
IOC	Initial Operating Capability
IPMDAR	Integrated Program Management Data and Analysis Report
IPMR	Integrated Program Management Report
IPT	Integrated Product Team
IR	Infrared
IRD	Interface Requirement Document
iRec	Internal Receivable
ISAR	Instrument Safety Assessment Report
ISRR	Instrument System Readiness Review
IT	Integration and Test
ITAR	International Traffic in Arms Regulations
JCT	Joint Compatibility Test
JPEG	Joint Photographic Experts Group
JPSS	Joint Polar satellite System
JSON	JavaScript Object Notation
K	Kelvin
KDP	Key Decision Point
Kg	kilograms
L	Launch
LEO	Low Earth Orbit
LEO&A	Launch, Early Orbit, and Activation
LOA	Letter of Agreement
LOE	Level of Effort
LPD	Low Earth Orbit Programs Division
LPT	Limited Performance Test
LRD	Launch Readiness Date
LRR	Launch Readiness Review
M&P	Materials and Processes
MA	Mission Assurance
MATLAB	Matrix Laboratory
Mbps	Mega bits per second
MBSE	Model Based Systems Engineering
MDR	Mission Definition Review
MGC	ASCII file format extension
MGSE	Mechanical Ground Support Equipment
MICD	Mechanical Interface Control Document
MIUL	Materials Identification and Usage List
MLI	Multi-Layer Insulation
MOC	Mission Operations Center
MOI	Moment of Inertia
MOR	Mission Operations Review
MOT	Mission Operations Team
MOTS	Modified-off-the-shelf
MOU	Memorandum of Understanding
MPEG	Moving Picture Experts Group

MPSCIP	Materials and Processes Selection, Control, and Implementation Plan
MR	Mission Rehearsal
MRB	Material Review Board
MRR	Manufacturing Readiness Review or Mission Readiness Review
MST	Mission Support Team
MUA	Materials Usage Agreement
N/A	Not Applicable
NASA	National Aeronautics and Space Administration
NASTRAN	NASA Structural Analysis
NEdT	Noise Equivalent Differential/Delta Temperature
NEON	Near Earth Orbit Network
NF	Noise Factor
NFS	NASA FAR Supplement
NGS	NOAA Ground System
NIST	National Institute of Standards and Technology
NITS	NASA IPMDAR Tailoring Supplement
NLT	No Later Than
NOAA	National Oceanic and Atmospheric Administration
NPR	NASA Procedural Requirement
NPSD	Noise Power Spectral Density
NS1	NEON Series-1 Project
NSOF	NOAA satellite Operations Facility
OBS	Organizational Breakdown Structure
ODAR	Orbital Debris Assessment Report
OHA	Operations Hazard Analysis
OHR	Operations Handover Review
OO	On-Orbit
ORR	Operational Readiness Review
P _{CS}	Probability of not meeting cost/schedule allocations
P _S	Probability of safety event occurrences
P _T	Probability of not meeting technical performance requirements
PADS	ASCII file format from Mentor Graphics
PCA	Physical Configuration Audit
PCB	Printed Circuit Board
PCM	Project Control Milestone
PDF	Portable Data Format
PDR	Preliminary Design Review
PDS	Post-Delivery Support
PER	Pre-Environmental Review
PHS&T	Packaging, Handling, Storage, and Transportation
PLAR	Post-Launch Assessment Review
PLD	Programmable Logic Device
PLO	Phase-Locked Oscillator
PMB	Performance Measurement Baseline
PM&P	Parts, Materials and Processes
PMP	Project Management Plan

PMPCA	Parts, Materials and Process Control Authority
PMR	Program Management Review
PPP	Part Production Plan
PRD	Performance Requirements Document
PROC	Procedures
PROD	Production
PROM	Programmable Only Memory
PRT	Platinum Resistance Thermometer
PSR	Instrument Pre-Ship Review
PSR	Program Status Review
QA	Quality Assurance
QPMR	Quarterly Program Management Review
R	For Review
RAM	Random Access Memory
RDR	Raw Data Record
RE	Review
Rec	Receivable
Rec-Del	Receive-Deliver
RF	Radio Frequency
RFE	Receiver Front End
RFA	Request for Action
RFI	Radio Frequency Interference
RHA	Radiation Hardness Assurance
RMM	Radiometric Math Model
RMP	Risk Management Plan
ROC	Receiver Operating Characteristic
ROP	Recommended Operating Procedures
RPS	Receiver Power Supply
RTM	Reduced Thermal Model
RVM	Requirement Verification Matrix
SC	Spacecraft
SDD	Software Detailed Design
SDE	Scan Drive Electronics
SDP	Software Development Plan
SDR	Sensor Data Record
SDR	System Definition Review
SE	Systems Engine
SEE	Single Event Effect
SEMP	Systems Engineering Management Plan
SEN	Software Engineering Notebook
SET	Single Event Transient
SFTP	Secure File Transfer Protocol
SIR	System Integration Review
SMA	Safety and Mission Assurance
SMBA	Sounder for Microwave-Based Applications
SOE	Satellite Operations Exercise
SOW	Statement of Work

SPA	Signal Processor Assembly
SPD	Schedule Performance Dataset
SPF	Single Point Failure
SPMM	Sensor Performance Math Model
SPVM	System Performance Verification Matrix
SPVP	System Performance Verification Plan
SRA	Schedule Risk Assessment
SRF	Spectral Response Function
SRR	Systems Requirements Review
SRR	Scene Release Reconstruction data file extension
STE	System Test Equipment
STEP	Standard for the Exchange of Product Data
SVT	Schedule Visibility Task
SW	Software
SWAR	Software Acceptance Review
SWE	Software Engineering Requirement
SWTRR	Software Test Readiness Review
T	Time of the launch date
TAA	Technical Assistance Agreement
TBD	To Be Determined
TBR	To Be Resolved
TBS	To Be Scheduled
TDA	Test Data Analysis
TDR	Temperature Data Record
TID	Total Ionizing Dose
TM	Torque Margin
TMM	Thermal Math Model
TO	Task Order
TPM	Technical Performance Metric
TQCM	Thermal Quartz Crystal Microbalance
TRL	Technology Readiness Level
TRR	Test Readiness Review
TVAC	Thermal Vacuum
TYF	Test as You Fly
UID	User Identification
U.S.	United States
VDD	Version Description Document
VHDL	Very High-Speed Integrated Circuits Hardware Description Language
VTL	Verification Tracking Log
W	Watt
WBS	Work Breakdown Structure
WGVC	Working Group on Calibration and Validation
XML	Extensible Markup Language